Quantum Information Theory

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Preface

These are lecture notes for the course NMMB537 Quantum Information Theory, taught at Faculty of Mathematics and Physics, Charles University. The lecture notes are mostly inspired by the course NMAK14020U Quantum Information Theory, offered at University of Copenhagen.

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Chapter 1

Quantum States and Measurements

We start by describing a precise mathematical framework for quantum states.

1.1 Finite-dimensional Hilbert Spaces

Axiom 1 (Hilbert space)**.** To every quantum system we asocciate a Hilbert space $\mathcal{H}$.

We only consider finite-dimensional Hilbert spaces in this course. If $\dim(\mathcal{H}) = d$, then $\mathcal{H}$ is isomorphic to

$$\mathbb{C}^d = \left\{ \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_d \end{pmatrix} : \psi_i \in \mathbb{C} \right\}$$

with the standard inner product

$$\langle \psi \mid \phi \rangle = \sum_{i=1}^{d} \overline{\psi_i} \phi_i.$$

Bra-ket notation. We write $|\psi\rangle \in \mathcal{H}$ for any vector. By $\langle\psi| \in \mathcal{H}^\dagger$, we mean the linear map $\mathcal{H} \to \mathbb{C}$ defined by $\langle\psi|(|\phi\rangle) = \langle \psi \mid \phi \rangle$. In other words, $\langle\psi|$ is the row vector corresponding to the conjugate transpose of $|\psi\rangle$. We denote the standard basis of $\mathbb{C}^d$ by $|0\rangle, |1\rangle, \ldots, |d-1\rangle$. With coordinates numbered $1, \ldots, d$, the vector $|i\rangle$ has 1 in coordinate $i+1$ and 0 elsewhere.

Linear algebra notation. Every basis of a Hilbert space that we consider will be orthonormal. If there is no confusion, we will use the terms linear map and matrix interchangeably.

By $\mathrm{Lin}(\mathcal{H}, \mathcal{K})$, we denote the set of linear maps $\mathcal{H} \to \mathcal{K}$. When we fix the standard bases, we can express $M \in \mathrm{Lin}(\mathcal{H}, \mathcal{K})$ as

$$M = \sum_{i,j} M_{ij} |i\rangle\langle j|.$$

Note that $\langle i|M|j\rangle = M_{ij} \in \mathbb{C}$.

Exercise 1.1. Write $\sum_{i,j \in \{0,1\}} ij|i\rangle\langle j|$ as a matrix.

If $M \in \mathrm{Lin}(\mathcal{H}, \mathcal{H}) = \mathrm{Lin}(\mathcal{H})$, then the trace with respect to the standard basis is

$$\mathrm{tr}[M] = \sum_i M_{ii} = \sum_i \langle i|M|i\rangle.$$

An important fact about trace is that it is in fact independent of the basis. Recall also that $\mathrm{tr}[MN] = \mathrm{tr}[NM]$.

Here are some important types of matrices:

- $M \in \mathrm{Lin}(\mathcal{H})$ is *Hermitian* if $M^\dagger = M$.
- $P \in \mathrm{Lin}(\mathcal{H})$ is *positive* if $\langle\psi|P|\psi\rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$. $\mathrm{PSD}(\mathcal{H})$ is the class of all positive matrices. We also use the notation $P \geq 0$ to indicate that P is positive. If $M, N \in \mathrm{Lin}(\mathcal{H})$, then we define $M \geq N$ if $M - N \geq 0$, that is, if $M - N$ is positive.
- $U \in \mathrm{Lin}(\mathcal{H})$ is *unitary* if $U^\dagger U = UU^\dagger = I$. $\mathcal{U}(\mathcal{H})$ is the class of all unitary matrices.
- $V \in \mathrm{Lin}(\mathcal{H}, \mathcal{K})$ is an *isometry* if $V^\dagger V = I$.
- $P \in \mathrm{Lin}(\mathcal{H})$ is an *orthogonal projection* if $P^\dagger = P$ and $P^2 = P$.

Exercise 1.2 (Why isometries preserve geometry)**.** Show that $V \in \mathrm{Lin}(\mathcal{H}, \mathcal{K})$ satisfies $V^\dagger V = I$ if and only if it preserves inner products. Deduce that it preserves lengths and distances.

Exercise 1.3 (Why these operators are orthogonal projections)**.** Show that the conditions $P^2 = P$ and $P^\dagger = P$ mean that P is the orthogonal projection onto its image $W = \mathrm{ran}(P)$.

Exercise 1.4. Are the following matrices Hermitian, PSD, unitary, or orthogonal projections?

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}.$$

Theorem 1.5 (Spectral theorem for Hermitian matrices)**.** *If $M \in \mathrm{Lin}(\mathcal{H})$ is Hermitian $d \times d$ matrix, then M has eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ and there is an orthonormal basis of eigenvectors $\{|\psi_i\rangle\}_{i=1}^d$ such that*

$$M = \sum_{i=1}^{d} \lambda_i |\psi_i\rangle\langle\psi_i|.$$

Recall that the eigenvalues can have multiple occurences and the basis of eigenvectors may not be unique. Indeed, for instance,

$$I = \sum_{i=1}^{d} |\psi_i\rangle\langle\psi_i|$$

is true for any orthonormal basis $\{|\psi_i\rangle\}_{i=1}^d$.

This allows us to define functions of Hermitian matrices. If $f : \mathbb{R} \to \mathbb{R}$ is a function, then we can define

$$f(M) = \sum_{i=1}^{d} f(\lambda_i) |\psi_i\rangle\langle\psi_i|.$$

For instance, if $M \geq 0$, the same construction with $f(t) = \sqrt{t}$ on $[0, \infty)$ defines its positive square root:

$$\sqrt{M} = \sum_{i=1}^{d} \sqrt{\lambda_i} |\psi_i\rangle\langle\psi_i|.$$

Note that if P is an orthogonal projection, then the condition $P^2 = P$ implies that the eigenvalues are in $\{0, 1\}$. Thus, we can write $P = \sum_{i=1}^{r} |\psi_i\rangle\langle\psi_i|$. Here, $r = \operatorname{rank}(P)$ of the orthogonal projection P.

Theorem 1.6 (Characterization of positive matrices)**.** *If $P \in \operatorname{Lin}(\mathcal{H})$, then the following are equivalent:*

(a) P is positive, i.e., $\langle\psi|P|\psi\rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$.

(b) $P^\dagger = P$ and all eigenvalues are nonnegative.

(c) There is $M \in \operatorname{Lin}(\mathcal{H})$ such that $P = M^\dagger M$.

(d) $\operatorname{tr}[PQ] \geq 0$, for every $Q \in \operatorname{PSD}(\mathcal{H})$.

Exercise 1.7. Prove that if $P \in \operatorname{PSD}(\mathcal{H})$ and $M \in \operatorname{Lin}(\mathcal{H}, \mathcal{K})$, then $MPM^\dagger$ is positive.

1.2 Quantum States

Definition 1.8. A *density matrix* is a positive operator $\rho \in \operatorname{PSD}(\mathcal{H})$ with $\operatorname{tr}[\rho] = 1$. We put $\mathcal{S}(\mathcal{H}) = \{\rho \in \operatorname{PSD}(\mathcal{H}) : \operatorname{tr}[\rho] = 1\}$.

Axiom 2. The state of a quantum system with Hilbert space $\mathcal{H}$ is described by a density matrix ρ. We will refer to ρ as a quantum state.

Example 1.9 (classical states)**.** In probability theory we have a finite set of outcomes Σ with a probability distribution p such that $p(x) \geq 0$, for $x \in \Sigma$, and $\sum_{x\in\Sigma} p(x) = 1$. If $\mathcal{H}$ has an orthonormal basis $\{|x\rangle\}_{x\in\Sigma}$, then

$$\rho = \sum_{x\in\Sigma} p(x)|x\rangle\langle x|$$

is a density matrix. Conversely, any diagonal density matrix corresponds to a probability distribution.

Example 1.10 (pure states)**.** Let $|\psi\rangle \in \mathcal{H}$ be a unit vector. Then $\rho = |\psi\rangle\langle\psi|$ is a density matrix since $\operatorname{tr}[|\psi\rangle\langle\psi|] = \operatorname{tr}[\langle\psi \mid \psi\rangle] = 1$. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$|\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\overline{\beta} \\ \overline{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

Note that replacing $|\psi\rangle$ by $e^{i\theta}|\psi\rangle$ does not change the density matrix.

Exercise 1.11. Is $\rho = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ a pure state?

Solution. Yes, since $\rho = |+\rangle\langle+|$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. □

Example 1.12 (general state)**.** Let ρ be a general quantum state. By Theorem 1.6, we can write

$$\rho = \sum_{i=1}^{d} p_i |\psi_i\rangle\langle\psi_i|,$$

where $p_i \geq 0$. Since ρ is diagonal in the basis $\{|\psi_i\rangle\}_{i=1}^d$ and trace is independent of the chosen basis, $\mathrm{tr}[\rho] = 1$ implies that $\sum_{i=1}^d p_i = 1$. So, the eigenvalues p_i's define a probability distribution. We may interpret ρ as if we have the pure state $|\psi_i\rangle\langle\psi_i|$ with probability p_i. We call ρ a *mixed state* if it is not pure.
If the spectrum is nondegenerate, this spectral decomposition is unique up to ordering and phases of the eigenvectors. Nevertheless, a mixed state admits other *ensemble decompositions* into pure states that need not be orthogonal, with different probabilities and possibly different numbers of states.

Exercise 1.13. What is the state corresponding to 50% chance of $|0\rangle$ and 50% chance of $|+\rangle$?

Solution.

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Note that the eigenvalues of ρ are $1/2 + \sqrt{2}/4$ and $1/2 - \sqrt{2}/4$, so the spectral decomposition of ρ gives a completely different decomposition of ρ compared to the one we started with. □

Definition 1.14. The state $\tau = \frac{1}{d}I$ is called the *maximally mixed state*.

Exercise 1.15 (Different preparations of the same state)**.** Compare the state obtained by choosing $|0\rangle$ or $|1\rangle$ with equal probabilities with the state obtained by choosing $|+\rangle$ or $|-\rangle$ with equal probabilities, where $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Solution. The first preparation gives

$$\rho_Z = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

For the second preparation, the off-diagonal entries cancel:

$$\rho_X = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{I}{2}.$$

Thus both preparations give the same maximally mixed state. In contrast, preparing the superposition $|+\rangle$ every time gives $|+\rangle\langle +| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Adding state vectors to form a superposition and averaging density matrices to describe a random preparation are different operations. □

1.3 Bloch Sphere

How does the set $\mathcal{S}(\mathcal{H})$ look like? For two states $\rho_1, \rho_2 \in \mathcal{S}(\mathcal{H})$ and $t \in [0, 1]$, the convex combination $t\rho_1 + (1-t)\rho_2$ is also a state. In fact, we have the following lemma.

Lemma 1.16. *The set $\mathcal{S}(\mathcal{H}) \subseteq \mathrm{Lin}(\mathcal{H})$ is convex. Moreoever, the extreme points of $\mathcal{S}(\mathcal{H})$ are exactly the pure states.*

For the particular case of one qubit, there is a visualization known as the Bloch sphere. Hermitian 2×2 matrices have the following (real) basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The last three matrices are known as Pauli matrices. Here are some of their properties:

$$X^2 = Y^2 = Z^2 = I, \qquad XY = iZ = -YX, \qquad \mathrm{tr}[X] = \mathrm{tr}[Y] = \mathrm{tr}[Z] = 0.$$

In particular, Pauli matrices are unitary and have eigenvalues ± 1.

Every 2×2 Hermitian matrix can be written in a unique way as a linear combination of the matrices I, X, Y, Z with all coefficients being real numbers. An arbitrary 2×2 Hermitian matrix ρ with $\mathrm{tr}[\rho] = 1$ can be expressed as

$$\rho = \frac{1}{2}(I + xX + yY + zZ) = \frac{1}{2}\begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}. \tag{1.1}$$

The coefficient $1/2$ in front of the I is fixed by the condition $\mathrm{tr}[\rho] = 1$. Further, $\mathrm{tr}[\rho] = 1$ implies that the eigenvalues of ρ are λ and $1 - \lambda$, for some $\lambda \in \mathbb{R}$. If we want ρ to be a quantum state, it also has to be positive, which is equivalent to requiring that $\lambda \geq 0$ and $1 - \lambda \geq 0$. Now, λ and $1 - \lambda$ are nonnegative if and only if $\det(\rho) = \lambda(1 - \lambda) \geq 0$. We compute

$$\det(\rho) = \frac{1}{4}((1+z)(1-z) - (x+iy)(x-iy)) = \frac{1}{4}(1 - x^2 - y^2 - z^2).$$

If we let $r = (x, y, z) \in \mathbb{R}^3$, then $\rho \geq 0$ if and only if $\|r\| \leq 1$. If ρ is a pure state, then $\mathrm{rank}(\rho) = 1$ and $\det(\rho) = 0$, which is equivalent to $\|r\| = 1$. Thus, pure states correspond exactly to the unit sphere in $\mathbb{R}^3$ and mixed states correspond to the open unit ball.

Exercise 1.17. What is the pure state $|\psi\rangle$ and its density matrix $|\psi\rangle\langle\psi|$ corresnponding to $r_1 = (0, 1, 0)$ and to $r_2 = (0, -1, 0)$.

Solution. By plugging r_1 into Eq. (1.1), we get

$$\rho_1 = \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

We can easily calculate, that the eigenvector corresnponding to the eigenvalue 1 of ρ_1 is $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$. By Theorem 1.5, we get $\rho_1 = |\psi_1\rangle\langle\psi_1|$. Similarly, we get that $\rho_2 = |\psi_2\rangle\langle\psi_2|$, where $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$. □

We have the following important bases of pure states that correspond to the intersection points of x, y, z-axes with the Bloch sphere:

- Z-basis: $\{|0\rangle, |1\rangle\}$, $r = (0, 0, 1)$, $-r = (0, 0, -1)$.
- X-basis: $\{|+\rangle, |-\rangle\}$, $r = (1, 0, 0)$, $-r = (-1, 0, 0)$
- Y-basis: $\{|+i\rangle, |-i\rangle\}$, $r = (0, 1, 0)$, $-r = (0, -1, 0)$.

Exercise 1.18 (Reading a Bloch vector)**.** Find the Bloch vector of the state $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13. Is it pure or mixed?

Solution. Comparing entries with Eq. (1.1), we obtain

$$\frac{1+z}{2} = \frac{3}{4}, \qquad \frac{x-iy}{2} = \frac{1}{4}, \qquad \text{so} \qquad r = \left(\frac{1}{2}, 0, \frac{1}{2}\right).$$

Its length is $\|r\| = 1/\sqrt{2} < 1$, so the state is mixed. The vector is also the average of the Bloch vectors of $|0\rangle$ and $|+\rangle$:

$$r = \frac{1}{2}(0,0,1) + \frac{1}{2}(1,0,0).$$

This illustrates how a probabilistic mixture becomes a convex combination of Bloch vectors. □

1.4 Measurements

Definition 1.19. A *measurement* or a *positive operator valued measure (POVM)* on a Hilbert space $\mathcal{H}$ with a finite set of outcomes Ω is a function $\mu : \Omega \to \mathrm{PSD}(\mathcal{H})$ such that $\sum_{x\in\Omega} \mu(x) = I$. If $\mu(x)$ are orthogonal projections, then the measurement is called *projective*.

Axiom 3. If we measure a state ρ using μ, then the probability of getting an outcome $x \in \Omega$ is $p(x) = \mathrm{tr}[\mu(x)\rho]$.

Remark 1.20. In general in quantum mechanics, it is not possible to perform measurements on a state without changing it. However, for now, we will not consider what happens with a state after the measurement (we can think that it is destroyed). We only get classical information about which outcome occured. We will discuss post-measurement states later in the course.

Exercise 1.21. Prove that if ρ is a quantum state and μ is a measurement, then $p(x) = \mathrm{tr}[\mu(x)\rho]$ defines a probability distribution on Ω.

Solution. By Theorem 1.6(d), we have $\mathrm{tr}[\mu(x)\rho] \geq 0$ since both $\mu(x)$ and ρ are positive. By the linearity of trace,

$$\sum_{x\in\Omega} \mathrm{tr}[\mu(x)\rho] = \mathrm{tr}\left[\sum_{x\in\Omega} \mu(x)\rho\right] = \mathrm{tr}[I\rho] = 1.$$

□

Example 1.22 (basis measurement)**.** Let $\{|\psi_i\rangle\}_{i=1}^d$ be an orthonormal basis and let $\mu(i) = |\psi_i\rangle\langle\psi_i|$ be the orthogonal projection onto $|\psi_i\rangle$. We get

$$p(i) = \mathrm{tr}[|\psi_i\rangle\langle\psi_i|\rho] = \mathrm{tr}[\langle\psi_i|\rho|\psi_i\rangle] = \langle\psi_i|\rho|\psi_i\rangle.$$

In particular, if $\rho = |\phi\rangle\langle\phi|$ is a pure state, then we get

$$p(i) = \langle\psi_i \mid \phi\rangle\langle\phi \mid \psi_i\rangle = |\langle\psi_i \mid \phi\rangle|^2.$$

Exercise 1.23. What are the outcome probabilities when we measure $|0\rangle$ the X-basis?

Solution. We get $|+\rangle$ with probability $|\langle + \mid 0\rangle|^2 = 1/2$ and $|-\rangle$ with probability $|\langle - \mid 0\rangle|^2 = 1/2$. □

Exercise 1.24 (Comparing two states by measurement)**.** Let $\rho = |+\rangle\langle +|$ and $\tau = I/2$. Calculate the outcome probabilities for each state when measuring in the Z-basis and in the X-basis.

Solution. For a Z-basis measurement, the probabilities are the diagonal entries in the standard basis. Both states have diagonal entries $1/2, 1/2$. For the X-basis measurement, the probability of outcome $+$ is

$$\langle +|\rho|+\rangle = |\langle + \mid +\rangle|^2 = 1, \qquad \langle +|\tau|+\rangle = \frac{1}{2}\langle + \mid +\rangle = \frac{1}{2}.$$

The probability of outcome $-$ is the complement. Altogether,

state	$p_Z(0)$	$p_Z(1)$	$p_X(+)$	$p_X(-)$
ρ	$1/2$	$1/2$	1	0
τ	$1/2$	$1/2$	$1/2$	$1/2$

The two states give the same statistics in the Z-basis, but different statistics in the X-basis. Knowing the probabilities for just one measurement need not determine the state. □

Exercise 1.25 (A nonprojective measurement)**.** Show that

$$\mu(0) = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad \mu(1) = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}$$

define a POVM that is not projective. What are the outcome probabilities on the state $|0\rangle$?

Solution. Both matrices are positive because they are diagonal with nonnegative entries, and $\mu(0) + \mu(1) = I$. They therefore define a POVM. However,

$$\mu(0)^2 = \begin{pmatrix} 9/16 & 0 \\ 0 & 1/16 \end{pmatrix} \neq \mu(0),$$

so the measurement is not projective. The Born rule gives

$$p(0) = \mathrm{tr}[\mu(0)|0\rangle\langle 0|] = \langle 0|\mu(0)|0\rangle = \frac{3}{4}, \qquad p(1) = \frac{1}{4}.$$

One way to obtain these measurement statistics is to measure in the Z-basis and then flip the reported outcome with probability $1/4$. □

1.5 Bloch Sphere Revisited

Given $r = (x, y, z)$ with $\|r\| = 1$, the positive matrices

$$\mu(0) := \rho(r) = \frac{1}{2}\begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} \quad \text{and} \quad \mu(1) := \rho(-r) = \frac{1}{2}\begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix}$$

define a general qubit basis measurement. Performing this measurement on a quantum state with Bloch vector $s = (x', y', z')$, then the probability of obtaining the outcome 0 is

$$p(0) = \frac{1}{2} + \frac{1}{2}(r \cdot s) = \frac{1}{2} + \frac{1}{2}(xx' + yy' + zz').$$

Geometrically $r \cdot s$ is the projection of the Bloch vector s of a quantum state onto the axis defining the measurment.

1.6 Observables

Sometimes it is convenient to reformulate measurements in terms of observables. A projective measurement is a measurement where all measurement operators are orthogonal projections, that is, we have a set of outcomes Ω and orthogonal projections $\{P_x : x \in \Omega\}$ such that $\sum_{x\in\Omega} P_x = I$. Suppose that $\Omega \subseteq \mathbb{R}$. The *observable* associated to the projective measurment is an operator $O \in \mathrm{Lin}(\mathcal{H})$ given by

$$O = \sum_{x\in\Omega} xP_x.$$

Since x are real, O is Hermitain, and, conversely, each Hermitian operator O has a spectral decomposition of this form, thereby defining a projective measurement.

Remark 1.26. In physics, observables are often the preferred way to reason about measurements. Measuring the observable O is the same as performing the projective measurement defined by the spectral decomposition of O.

Example 1.27. Since $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, the observable Z corresponds to measuring in the Z-basis with outcomes $\{\pm 1\}$. Similarly, since $X = |+\rangle\langle +| - |-\rangle\langle -|$, measuring X corresponds to measuring in the X-basis with outcomes $\{\pm 1\}$. More generally, we may measure a qubit in the basis given by the antipodal points $r = (x, y, z)$ and $-r$ on the Bloch sphere with outcomes $\{\pm 1\}$. This gives an observable

$$\begin{aligned} O(r) = \rho(r) - \rho(-r) &= \frac{1}{2}\begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix} \\ &= \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}. \end{aligned}$$

States and expectation values. In classical probability, a probability distribution p on a finite set Σ assigns an expectation to each real-valued random variable $a : \Sigma \to \mathbb{R}$:

$$\mathbb{E}_p[a] = \sum_{s\in\Sigma} p(s)a(s).$$

A quantum state plays a similar role: it assigns an expectation value to each observable. The observable specifies the measurement and its numerical outcomes, while the state determines their probabilities.

Let $O = \sum_{x\in\Omega} xP_x$ and let the state be ρ. The measurement outcome is an ordinary real-valued random variable, with $p_\rho(x) = \mathrm{tr}[P_x\rho]$ by the Born rule. We write $\mathbb{E}_\rho[O]$ for its expectation. Taking the weighted average of the outcomes and using linearity of the trace gives

$$\begin{aligned}\mathbb{E}_\rho[O] &= \sum_{x\in\Omega} x\, p_\rho(x) = \sum_{x\in\Omega} x\ \mathrm{tr}[P_x\rho] \\ &= \mathrm{tr}\left[\left(\sum_{x\in\Omega} xP_x\right)\rho\right] = \mathrm{tr}[O\rho] = \mathrm{tr}[\rho O].\end{aligned}$$

For a pure state $\rho = |\psi\rangle\langle\psi|$, this becomes

$$\mathbb{E}_\rho[O] = \mathrm{tr}[O|\psi\rangle\langle\psi|] = \langle\psi|O|\psi\rangle.$$

Thus, we can view a state as an *expectation functional* $O \mapsto \mathbb{E}_\rho[O]$. Just like classical expectation, it is linear, normalized, and positive: for observables A, B and $a, b \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}_\rho[aA + bB] &= a\mathbb{E}_\rho[A] + b\mathbb{E}_\rho[B], \\ \mathbb{E}_\rho[I] &= 1, \\ A \geq 0 \implies \mathbb{E}_\rho[A] &\geq 0.\end{aligned}$$

The first two properties follow from linearity of trace and $\mathrm{tr}[\rho] = 1$; the last follows from Theorem 1.6(d). Each expectation concerns the corresponding measurement on identically prepared systems.

Probabilities as expectations. An orthogonal projection P is itself an observable with possible outcomes in $\{0, 1\}$. It describes a yes-or-no question, and its expectation is the probability of the answer 1. In this sense, orthogonal projections play the role of indicator random variables. In particular, the probability of outcome x when measuring O is

$$p_\rho(x) = \mathrm{tr}[\rho P_x] = \mathbb{E}_\rho[P_x].$$

The expectation functional therefore determines the entire outcome distribution of every projective measurement. More generally, the probability of outcome x for any POVM μ is $\mathbb{E}_\rho[\mu(x)]$.

Random preparations also obey the usual law of total expectation. If state ρ_i is prepared with probability p_i, then the average state is $\rho = \sum_i p_i\rho_i$ and

$$\mathbb{E}_\rho[O] = \mathrm{tr}\left[\left(\sum_i p_i\rho_i\right)O\right] = \sum_i p_i\, \mathbb{E}_{\rho_i}[O].$$

For $O = \mu(x)$, the same identity is precisely the law of total probability for the measurement outcome x.

Exercise 1.28 (An expectation value is an average)**.** For $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13, calculate the expectation value of Z using both the outcome probabilities and the trace formula.

Solution. The outcomes $+1$ and -1 correspond to $|0\rangle$ and $|1\rangle$, respectively. Their probabilities are $3/4$ and $1/4$, so

$$\mathbb{E}(\text{outcome}) = (+1)\frac{3}{4} + (-1)\frac{1}{4} = \frac{1}{2}.$$

The trace formula gives the same result:

$$\operatorname{tr}[Z\rho] = \operatorname{tr}\begin{pmatrix} 3/4 & 1/4 \\ -1/4 & -1/4 \end{pmatrix} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

A single measurement returns $+1$ or -1. The expectation value $1/2$ describes the average over many repetitions on identically prepared systems. □

Chapter 2

Multiple Quantum Systems

2.1 Multiple Systems

If classical random variables have outcome sets $\Sigma_1, \ldots, \Sigma_n$, their joint distribution is a probability distribution on the product set

$$\Sigma_1 \times \cdots \times \Sigma_n = \{(x_1, \ldots, x_n) : x_j \in \Sigma_j\}.$$

In particular, the possible values of n bits are the tuples in $\{0,1\}^n$.

The tensor product plays the corresponding role for quantum systems. We first define it for two finite-dimensional Hilbert spaces.

Tensor product of Hilbert spaces. For Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, start with finite formal linear combinations of symbols $|\psi\rangle \otimes |\phi\rangle$, where $|\psi\rangle \in \mathcal{H}$ and $|\phi\rangle \in \mathcal{K}$. We identify such sums using the following *bilinearity relations*, for $\alpha, \beta \in \mathbb{C}$:

$$(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) \otimes |\phi\rangle = \alpha(|\psi_1\rangle \otimes |\phi\rangle) + \beta(|\psi_2\rangle \otimes |\phi\rangle),$$
$$|\psi\rangle \otimes (\alpha|\phi_1\rangle + \beta|\phi_2\rangle) = \alpha(|\psi\rangle \otimes |\phi_1\rangle) + \beta(|\psi\rangle \otimes |\phi_2\rangle).$$

The resulting vector space is the *tensor product* $\mathcal{H} \otimes \mathcal{K}$. Its elements are sums of product vectors; they need not themselves be a single product vector. We equip it with the inner product determined by

$$\langle \psi \otimes \phi, \psi' \otimes \phi' \rangle = \langle \psi \mid \psi' \rangle \langle \phi \mid \phi' \rangle,$$

extended conjugate-linearly in the first argument and linearly in the second. The bilinearity relations ensure that this is well-defined.

Product bases. Choose orthonormal bases $\{|a\rangle\}$ of $\mathcal{H}$ and $\{|b\rangle\}$ of $\mathcal{K}$, and abbreviate $|a\rangle \otimes |b\rangle$ as $|ab\rangle$. Bilinearity gives

$$\left(\sum_a \psi_a |a\rangle\right) \otimes \left(\sum_b \phi_b |b\rangle\right) = \sum_{a,b} \psi_a \phi_b |ab\rangle.$$

Thus the vectors $|ab\rangle$ span $\mathcal{H} \otimes \mathcal{K}$. They satisfy $\langle ab \mid a'b' \rangle = \delta_{aa'}\delta_{bb'}$, so they form an orthonormal basis. A general vector has the form $|\Psi\rangle = \sum_{a,b} c_{ab}|ab\rangle$; for a single product vector, its coefficients factor as $c_{ab} = \psi_a \phi_b$.

For more factors, we repeat this construction and identify different bracketings by $(u \otimes v) \otimes w \leftrightarrow u \otimes (v \otimes w)$. If $\{|x_j\rangle : x_j \in \Sigma_j\}$ is an orthonormal basis of $\mathcal{H}_j$, then $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$ has the orthonormal basis

$$\{|x_1\rangle \otimes \cdots \otimes |x_n\rangle : x_j \in \Sigma_j\}.$$

We also write $|x_1\rangle \cdots |x_n\rangle$ or $|x_1 \cdots x_n\rangle$. Counting the basis vectors gives

$$\dim(\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n) = \dim(\mathcal{H}_1) \cdots \dim(\mathcal{H}_n).$$

Axiom 4 (Composite systems)**.** If quantum systems have Hilbert spaces $\mathcal{H}_1, \ldots, \mathcal{H}_n$, then their joint system has Hilbert space $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$.

Example 2.1. For two qubits the joint Hilbert space is $\mathbb{C}^2 \otimes \mathbb{C}^2$. It is four-dimensional, with standard basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, in this order. For n qubits the Hilbert space is $(\mathbb{C}^2)^{\otimes n}$, with dimension 2^n and standard basis $\{|x_1 \cdots x_n\rangle : x_1, \ldots, x_n \in \{0, 1\}\}$.

Notation. We label systems by $A, B, C, \ldots$ and their Hilbert spaces by $\mathcal{H}_A, \mathcal{H}_B, \mathcal{H}_C, \ldots$. We write $|A| = \dim \mathcal{H}_A$, $\mathrm{Lin}(A) = \mathrm{Lin}(\mathcal{H}_A)$, $\mathcal{S}(A) = \mathcal{S}(\mathcal{H}_A)$, and $\mathrm{PSD}(A) = \mathrm{PSD}(\mathcal{H}_A)$. The label AB denotes the composite system with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$. Thus $\rho_{AB} \in \mathcal{S}(AB)$ may describe a state shared by Alice and Bob, while μ_A is a measurement on Alice's system. Tensor factors are always written in the order of the subsystem labels.

Tensor products of operators. For $M_A \in \mathrm{Lin}(A)$ and $N_B \in \mathrm{Lin}(B)$, the operator $M_A \otimes N_B \in \mathrm{Lin}(AB)$ is determined by

$$(M_A \otimes N_B)(|\psi_A\rangle \otimes |\phi_B\rangle) = (M_A|\psi_A\rangle) \otimes (N_B|\phi_B\rangle)$$

and linearity. The same definition applies to maps between different input and output spaces. In product bases its entries are

$$(M_A \otimes N_B)_{ab,a'b'} = (M_A)_{a,a'}(N_B)_{b,b'}.$$

This is the Kronecker product of matrices: each entry of M_A is replaced by that entry times the matrix N_B.

Example 2.2. If $M_A = |a\rangle\langle a'|$ and $N_B = |b\rangle\langle b'|$, then

$$M_A \otimes N_B = |a\rangle\langle a'| \otimes |b\rangle\langle b'| = |ab\rangle\langle a'b'|.$$

Definition 2.3. A state of the form $\rho_{AB} = \rho_A \otimes \rho_B$, where $\rho_A \in \mathcal{S}(A)$ and $\rho_B \in \mathcal{S}(B)$, is a *product state*. A state that is not a product state is called *correlated*.

This is a density matrix: the tensor product of positive operators is positive, as follows by taking their spectral decompositions, and

$$\mathrm{tr}[M_A \otimes N_B] = \mathrm{tr}[M_A]\,\mathrm{tr}[N_B].$$

In particular, $\mathrm{tr}[\rho_A \otimes \rho_B] = 1$.

Example 2.4 (Classical states)**.** Let p_{XY} be a joint probability distribution on $\Sigma_X \times \Sigma_Y$. Its density matrix is

$$\rho_{XY} = \sum_{x,y} p_{XY}(x,y)|xy\rangle\langle xy| = \sum_{x,y} p_{XY}(x,y)|x\rangle\langle x| \otimes |y\rangle\langle y|.$$

Conversely, every density matrix diagonal in this product basis has this form. The state ρ_{XY} is a product state if and only if X and Y are independent under p_{XY}. Product states therefore generalize independence in classical probability.

Example 2.5 (Perfect classical correlation)**.** The classical state

$$\sigma_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|) = \frac{1}{2}(|0\rangle\langle 0| \otimes |0\rangle\langle 0| + |1\rangle\langle 1| \otimes |1\rangle\langle 1|)$$

describes two equal, uniformly random bits. It is not a product state: each bit is uniform, but the outcomes 01 and 10 have probability zero. In the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\sigma_{AB} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Exercise 2.6. Let

$$\rho_A = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \rho_B = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Express $\rho_A \otimes \rho_B$ in the standard basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Solution. Using the block form of the Kronecker product, we obtain

$$\rho_A \otimes \rho_B = \frac{1}{2}\begin{pmatrix} \rho_B & \rho_B \\ \rho_B & \rho_B \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

□

Example 2.7 (Pure product states)**.** For two qubits, the basis vectors $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are pure product states. Another example is

$$|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

We freely abbreviate $|\psi_A\rangle \otimes |\phi_B\rangle$ as $|\psi_A\rangle|\phi_B\rangle$.

Example 2.8 (A maximally entangled state)**.** The Bell state

$$|\Phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

is a pure state that is not a product state. We will explain the term *maximally entangled* in

Section 2.5. Its density matrix is

$$\rho_{AB} = |\Phi^+_{AB}\rangle\langle\Phi^+_{AB}| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$$
$$= \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

For example, $|00\rangle\langle 11| = |0\rangle\langle 1| \otimes |0\rangle\langle 1|$. The off-diagonal entries distinguish this state from σ_{AB} in Example 2.5.

2.2 The Partial Trace

Classical correlations do not communicate. Suppose a fair coin is tossed and its result is written on two cards, one in Alice's sealed envelope and one in Bob's. Neither has seen the result. Write X for Alice's bit and Y for Bob's. This is the preparation described by the density matrix in Example 2.5. Its four diagonal entries are the *joint* probabilities for $00, 01, 10, 11$:

$$(p_{XY}(0,0), p_{XY}(0,1), p_{XY}(1,0), p_{XY}(1,1)) = \left(\frac{1}{2}, 0, 0, \frac{1}{2}\right).$$

Each individual bit is uniform, because we sum over the other bit:

$$p_X(0) = p_{XY}(0,0) + p_{XY}(0,1) = \frac{1}{2} + 0 = \frac{1}{2},$$

and similarly $p_X(1) = p_Y(0) = p_Y(1) = 1/2$. Bob travels far away and opens his envelope. If he sees $Y = 0$, he immediately knows that Alice's card also says 0. This is *conditioning*: $p_{X|Y}(0 \mid 0) = 1$. He has learned an outcome and updated his prediction about Alice's card. Alice, with her envelope still closed and without Bob's result, still assigns probability $1/2$ to each value of X. Her *marginal distribution* $p_X(x) = \sum_y p_{XY}(x,y)$ is unchanged whether Bob opens his envelope or leaves it closed. She cannot tell which he did, and he cannot choose the random bit he reveals. The correlation was established when the cards were prepared; opening an envelope sends no message.

The quantum no-communication statement below has the same distinction between conditional and marginal probabilities, and holds for every shared state and every choice of local measurement. Its proof makes no assumption that measurement outcomes were fixed in advance.

Local measurements. If Alice performs a measurement $\mu_A : \Omega_A \to \mathrm{PSD}(A)$, its description on the joint system is the POVM $x \mapsto \mu_A(x) \otimes I_B$. Indeed, its operators are positive and

$$\sum_{x\in\Omega_A} \mu_A(x) \otimes I_B = \left(\sum_{x\in\Omega_A} \mu_A(x)\right) \otimes I_B = I_A \otimes I_B.$$

If Bob also performs a measurement $\mu_B : \Omega_B \to \mathrm{PSD}(B)$, their joint measurement has effects $\mu_A(x) \otimes \mu_B(y)$ and probabilities

$$p(x,y) = \mathrm{tr}[(\mu_A(x) \otimes \mu_B(y))\rho_{AB}].$$

No-communication for local measurements. For fixed local measurements, $p(x, y)$ is an ordinary joint probability distribution. Alice's outcome distribution is its *marginal*, obtained by summing over Bob's outcome:

$$\begin{aligned} p_A(x) &= \sum_y p(x,y) = \sum_y \mathrm{tr}[(\mu_A(x) \otimes \mu_B(y))\rho_{AB}] \\ &= \mathrm{tr}\left[\left(\mu_A(x) \otimes \sum_y \mu_B(y)\right)\rho_{AB}\right] = \mathrm{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}]. \end{aligned}$$

Thus Alice's outcome distribution does not depend on Bob's choice of measurement: every POVM sums to I_B, so the same marginal results for every choice. The same holds with Alice and Bob interchanged. They cannot send a message merely by choosing a local measurement. Conditioning on an outcome known to Bob can change Alice's conditional statistics, but Alice needs to learn that outcome to use them. This argument concerns outcome probabilities; it requires no rule for the post-measurement state.

At the level of states, the quantum analogue of taking a marginal distribution is the *partial trace*. What state describes Alice's system alone? We seek ρ_A such that

$$\mathrm{tr}[\mu_A(x)\rho_A] = \mathrm{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}]$$

for every local measurement. Choose orthonormal bases $\{|a\rangle\}$ and $\{|b\rangle\}$ of $\mathcal{H}_A$ and $\mathcal{H}_B$. Since $|ab\rangle = (I_A \otimes |b\rangle)|a\rangle$ and $\langle ab| = \langle a|(I_A \otimes \langle b|)$, expanding the trace yields

$$\begin{aligned} \mathrm{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}] &= \sum_{a,b} \langle ab|(\mu_A(x) \otimes I_B)\rho_{AB}|ab\rangle \\ &= \sum_{a,b} \langle a|(I_A \otimes \langle b|)(\mu_A(x) \otimes I_B)\rho_{AB}(I_A \otimes |b\rangle)|a\rangle \\ &= \sum_{a,b} \langle a|\mu_A(x)(I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle)|a\rangle \\ &= \sum_a \langle a|\mu_A(x)\left(\sum_b (I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle)\right)|a\rangle. \end{aligned}$$

The third line uses

$$(I_A \otimes \langle b|)(\mu_A(x) \otimes I_B) = \mu_A(x) \otimes \langle b| = \mu_A(x)(I_A \otimes \langle b|).$$

Both equalities follow from the tensor-product multiplication rule; for the second, write $\mu_A(x) = \mu_A(x) \otimes [1]$, where $[1]$ is the 1×1 identity matrix. The last line is $\mathrm{tr}[\mu_A(x)\rho_A]$ if we set

$$\rho_A = \sum_b (I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle).$$

Two qubits in matrix notation. We can obtain the same result using ordinary matrix multiplication. In the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, write

$$\rho_{AB} = \left(\begin{array}{cc|cc} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ \hline r_{31} & r_{32} & r_{33} & r_{34} \\ r_{41} & r_{42} & r_{43} & r_{44} \end{array}\right).$$

For Alice's measurement effect $E = \mu_A(x)$, we have

$$E = \begin{pmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{pmatrix}, \qquad E \otimes I_2 = \begin{pmatrix} e_{11} & 0 & e_{12} & 0 \\ 0 & e_{11} & 0 & e_{12} \\ e_{21} & 0 & e_{22} & 0 \\ 0 & e_{21} & 0 & e_{22} \end{pmatrix}.$$

Multiplying the matrices and summing the diagonal entries gives

$$\begin{aligned} \operatorname{tr}[(E \otimes I_2)\rho_{AB}] = {} & e_{11}(r_{11} + r_{22}) + e_{12}(r_{31} + r_{42}) \\ & + e_{21}(r_{13} + r_{24}) + e_{22}(r_{33} + r_{44}). \end{aligned}$$

This is exactly $\operatorname{tr}[E\rho_A]$ for the 2×2 matrix

$$\rho_A = \begin{pmatrix} r_{11} + r_{22} & r_{13} + r_{24} \\ r_{31} + r_{42} & r_{33} + r_{44} \end{pmatrix}.$$

Thus each entry of ρ_A is the trace of the corresponding 2×2 block of ρ_{AB}. For example, its upper-right entry is

$$r_{13} + r_{24} = \langle 00|\rho_{AB}|10\rangle + \langle 01|\rho_{AB}|11\rangle = \sum_{b=0}^{1} \langle 0b|\rho_{AB}|1b\rangle.$$

This is precisely the sum over Bob's basis index in the bra-ket calculation. On the diagonal it gives the usual marginal probabilities for computational-basis measurements; the same summation also determines the off-diagonal entries.

Both calculations motivate the following definition.

Definition 2.9 (Partial trace)**.** For $M_{AB} \in \operatorname{Lin}(AB)$, its *partial traces over B and A* are

$$\begin{aligned} \operatorname{tr}_B[M_{AB}] &= \sum_b (I_A \otimes \langle b|) M_{AB} (I_A \otimes |b\rangle), \\ \operatorname{tr}_A[M_{AB}] &= \sum_a (\langle a| \otimes I_B) M_{AB} (|a\rangle \otimes I_B), \end{aligned}$$

where $\{|a\rangle\}$ and $\{|b\rangle\}$ are any orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$. For a state ρ_{AB}, the operators

$$\rho_A = \operatorname{tr}_B[\rho_{AB}], \qquad \rho_B = \operatorname{tr}_A[\rho_{AB}]$$

are called its *reduced states* on A and B, respectively. The subscript of tr names the system that is discarded.

We will verify shortly that these definitions are independent of the bases and that both reduced states are density matrices. Inserting the definitions between basis bras and kets of the remaining system gives their matrix entries:

$$\begin{aligned} \langle a| \operatorname{tr}_B[M_{AB}]|a'\rangle &= \sum_b \langle a|(I_A \otimes \langle b|) M_{AB} (I_A \otimes |b\rangle)|a'\rangle \\ &= \sum_b \langle ab|M_{AB}|a'b\rangle, \\ \langle b| \operatorname{tr}_A[M_{AB}]|b'\rangle &= \sum_a \langle b|(\langle a| \otimes I_B) M_{AB} (|a\rangle \otimes I_B)|b'\rangle \\ &= \sum_a \langle ab|M_{AB}|ab'\rangle. \end{aligned} \tag{2.1}$$

Equivalently, if

$$M_{AB} = \sum_{a,a',b,b'} M_{ab,a'b'} |a\rangle\langle a'| \otimes |b\rangle\langle b'|,$$

then reconstructing the operator from these matrix entries yields

$$\begin{aligned} \mathrm{tr}_B[M_{AB}] &= \sum_{a,a'} \langle a| \mathrm{tr}_B[M_{AB}] |a'\rangle \, |a\rangle\langle a'| \\ &= \sum_{a,a'} \left(\sum_b M_{ab,a'b} \right) |a\rangle\langle a'|, \\ \mathrm{tr}_A[M_{AB}] &= \sum_{b,b'} \langle b| \mathrm{tr}_A[M_{AB}] |b'\rangle \, |b\rangle\langle b'| \\ &= \sum_{b,b'} \left(\sum_a M_{ab,ab'} \right) |b\rangle\langle b'|. \end{aligned}$$

For tr_B, only entries with the same B index in the row and column contribute; for tr_A, the A indices must agree. For a tensor product, the same contraction separates into its two factors:

$$\begin{aligned} \mathrm{tr}_B[N_A \otimes O_B] &= \sum_b (I_A \otimes \langle b|)(N_A \otimes O_B)(I_A \otimes |b\rangle) \\ &= \sum_b N_A \langle b|O_B|b\rangle = N_A \, \mathrm{tr}[O_B], \\ \mathrm{tr}_A[N_A \otimes O_B] &= \sum_a (\langle a| \otimes I_B)(N_A \otimes O_B)(|a\rangle \otimes I_B) \\ &= \sum_a \langle a|N_A|a\rangle \, O_B = \mathrm{tr}[N_A] \, O_B. \end{aligned} \tag{2.2}$$

Every operator is a linear combination of tensor products of matrix units, so these rules determine both partial traces on all operators. They also show that the reduced states of a product state $\rho_A \otimes \rho_B$ are precisely ρ_A and ρ_B, since both factors have trace one.

Lemma 2.10 (Properties of the partial trace)**.**

(a) The maps $\mathrm{tr}_B : \mathrm{Lin}(AB) \to \mathrm{Lin}(A)$ *and* $\mathrm{tr}_A : \mathrm{Lin}(AB) \to \mathrm{Lin}(B)$ *are linear.*

(b) For $N_A \in \mathrm{Lin}(A)$*,* $O_B \in \mathrm{Lin}(B)$*, and* $M_{AB} \in \mathrm{Lin}(AB)$*,*

$$\begin{aligned} \mathrm{tr}[(N_A \otimes I_B) M_{AB}] &= \mathrm{tr}[N_A \, \mathrm{tr}_B[M_{AB}]], \\ \mathrm{tr}[(I_A \otimes O_B) M_{AB}] &= \mathrm{tr}[O_B \, \mathrm{tr}_A[M_{AB}]]. \end{aligned}$$

(c) Neither partial trace depends on the orthonormal basis of the system being traced out.

(d) $\mathrm{tr}[\mathrm{tr}_B[M_{AB}]] = \mathrm{tr}[M_{AB}] = \mathrm{tr}[\mathrm{tr}_A[M_{AB}]]$.

(e) If $P_{AB} \geq 0$*, then* $\mathrm{tr}_B[P_{AB}] \geq 0$ *and* $\mathrm{tr}_A[P_{AB}] \geq 0$.

Proof. We first consider tr_B; the arguments for tr_A are the same with the roles of A and B interchanged.

(a) Linearity. Matrix multiplication distributes over sums, so substituting $\alpha M_{AB} + \beta N_{AB}$ into the defining formula gives

$$\operatorname{tr}_B[\alpha M_{AB} + \beta N_{AB}] = \alpha \operatorname{tr}_B[M_{AB}] + \beta \operatorname{tr}_B[N_{AB}].$$

(b) Local trace identity. The calculation preceding Definition 2.9 applies with N_A in place of $\mu_A(x)$ and M_{AB} in place of ρ_{AB}. It uses only the rules for traces and tensor products, so N_A and M_{AB} need not be positive or Hermitian. It also works for every orthonormal basis chosen in the defining sum.

(c) Independence of the basis. Fix M_{AB}, and let two orthonormal bases $\{|b\rangle\}$ and $\{|b'\rangle\}$ of $\mathcal{H}_B$ give the operators

$$T = \sum_b (I_A \otimes \langle b|) M_{AB} (I_A \otimes |b\rangle),$$
$$T' = \sum_{b'} (I_A \otimes \langle b'|) M_{AB} (I_A \otimes |b'\rangle).$$

Applying (b) separately to these two sums gives, for every $N_A \in \operatorname{Lin}(A)$,

$$\operatorname{tr}[N_A T] = \operatorname{tr}[(N_A \otimes I_B) M_{AB}] = \operatorname{tr}[N_A T'].$$

The middle expression makes no reference to either basis of B. Thus $D = T - T'$ satisfies $\operatorname{tr}[N_A D] = 0$ for every N_A. To see why this forces $D = 0$, fix an orthonormal basis $\{|a\rangle\}$ of $\mathcal{H}_A$ and choose the matrix unit $N_A = |a'\rangle\langle a|$. Expanding the trace in that basis gives

$$\begin{aligned} 0 = \operatorname{tr}[|a'\rangle\langle a| D] &= \sum_c \langle c \mid a'\rangle \langle a|D|c\rangle \\ &= \sum_c \delta_{c,a'} \langle a|D|c\rangle = \langle a|D|a'\rangle. \end{aligned}$$

In other words, this choice of N_A extracts the (a, a') matrix entry of D. Since a, a' were arbitrary, every entry is zero, so $T = T'$. This proves basis independence for every M_{AB}. For tr_A, repeat the argument in a fixed orthonormal basis $\{|c\rangle\}$ of the remaining system B, using the matrix units $|c'\rangle\langle c|$.

(d) Preservation of the trace. Taking $N_A = I_A$ in (b) gives

$$\operatorname{tr}[\operatorname{tr}_B[M_{AB}]] = \operatorname{tr}[(I_A \otimes I_B) M_{AB}] = \operatorname{tr}[M_{AB}].$$

Taking $O_B = I_B$ in the second identity of (b) gives the same result for tr_A.

(e) Preservation of positivity. Recall that $P_{AB} \geq 0$ means that P_{AB} is Hermitian and $\langle v|P_{AB}|v\rangle \geq 0$ for every vector $|v\rangle$ of the joint system. We check these two properties for $\operatorname{tr}_B[P_{AB}]$. Set $V_b = I_A \otimes |b\rangle$, so that $V_b^\dagger = I_A \otimes \langle b|$. Each summand in the partial trace is Hermitian, because

$$(V_b^\dagger P_{AB} V_b)^\dagger = V_b^\dagger P_{AB}^\dagger V_b = V_b^\dagger P_{AB} V_b.$$

Their sum $\operatorname{tr}_B[P_{AB}]$ is therefore Hermitian as well.

Now take any $|\psi\rangle \in \mathcal{H}_A$. The vector $|w_b\rangle = V_b|\psi\rangle = |\psi\rangle \otimes |b\rangle$ belongs to $\mathcal{H}_A \otimes \mathcal{H}_B$, so positivity of P_{AB} gives $\langle w_b|P_{AB}|w_b\rangle \geq 0$ for each b. Consequently,

$$\begin{aligned} \langle\psi| \operatorname{tr}_B[P_{AB}] |\psi\rangle &= \sum_b \langle\psi|V_b^\dagger P_{AB} V_b|\psi\rangle \\ &= \sum_b (\langle\psi| \otimes \langle b|) P_{AB} (|\psi\rangle \otimes |b\rangle) \\ &= \sum_b \langle w_b|P_{AB}|w_b\rangle \geq 0. \end{aligned}$$

Every term is nonnegative, and this holds for every $|\psi\rangle$; hence $\operatorname{tr}_B[P_{AB}] \geq 0$. Similarly, for every $|\phi\rangle \in \mathcal{H}_B$,

$$\langle\phi| \operatorname{tr}_A[P_{AB}]|\phi\rangle = \sum_a (\langle a| \otimes \langle\phi|) P_{AB} (|a\rangle \otimes |\phi\rangle) \geq 0,$$

and the same Hermiticity argument gives $\operatorname{tr}_A[P_{AB}] \geq 0$. □

Parts (d) and (e) show that both reduced states are density matrices. Part (b) says that they reproduce every local expectation value:

$$\mathbb{E}_{\rho_A}[O_A] = \mathbb{E}_{\rho_{AB}}[O_A \otimes I_B],$$
$$\mathbb{E}_{\rho_B}[O_B] = \mathbb{E}_{\rho_{AB}}[I_A \otimes O_B].$$

Each is uniquely determined even by local measurement probabilities, since rank-one projections determine the quadratic form of a Hermitian operator.

Example 2.11 (Reduced states of a Bell pair)**.** For $\rho_{AB} = |\Phi^+_{AB}\rangle\langle\Phi^+_{AB}|$, Eq. (2.2) gives

$$\begin{aligned}\operatorname{tr}_B[\rho_{AB}] &= \frac{1}{2}(|0\rangle\langle 0| \operatorname{tr}[|0\rangle\langle 0|] + |0\rangle\langle 1| \operatorname{tr}[|0\rangle\langle 1|] \\ &\qquad + |1\rangle\langle 0| \operatorname{tr}[|1\rangle\langle 0|] + |1\rangle\langle 1| \operatorname{tr}[|1\rangle\langle 1|]) \\ &= \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I_A.\end{aligned}$$

Tracing out A instead gives

$$\rho_B = \operatorname{tr}_A[\rho_{AB}] = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I_B.$$

These are also the reduced states of the classically correlated state σ_{AB} and of the product state $(I_A/2) \otimes (I_B/2)$. Alice alone cannot distinguish these three joint preparations, because all her local measurement probabilities are identical.

Mixed states arise from probabilistic preparations and from restricting a joint state to a subsystem, even when the joint state is pure. A measurement also produces a classical outcome distribution, which can be represented by a diagonal density matrix on a classical register. This last state describes the recorded outcome, not a post-measurement state specified by the POVM alone.

Example 2.12 (Marginal distributions)**.** For the classical joint state

$$\rho_{XY} = \sum_{x,y} p_{XY}(x,y) |x\rangle\langle x| \otimes |y\rangle\langle y|,$$

the reduced states are

$$\rho_X = \operatorname{tr}_Y[\rho_{XY}] = \sum_x \left(\sum_y p_{XY}(x,y) \right) |x\rangle\langle x| = \sum_x p_X(x) |x\rangle\langle x|,$$

$$\rho_Y = \operatorname{tr}_X[\rho_{XY}] = \sum_y \left(\sum_x p_{XY}(x,y) \right) |y\rangle\langle y| = \sum_y p_Y(y) |y\rangle\langle y|.$$

Thus partial trace reduces to taking the marginal distributions $p_X(x) = \sum_y p_{XY}(x,y)$ and $p_Y(y) = \sum_x p_{XY}(x,y)$. In particular, $p_{XY}(x,y) = p_X(x) p_Y(y)$ for all x, y if and only if $\rho_{XY} = \rho_X \otimes \rho_Y$.

Classical conditional probabilities also make sense after a measurement outcome is specified. For general quantum systems, there is no single measurement-independent conditional density matrix obtained by simply dividing a joint state by a marginal as in classical probability.

Exercise 2.13 (Tensor products of observables)**.** Let $O_A = \sum_x xP_x$ and $O_B = \sum_y yQ_y$ be observables. Show that their local measurements define the joint projective measurement $\{P_x \otimes Q_y\}_{x,y}$ and that

$$\mathbb{E}[xy] = \operatorname{tr}[\rho_{AB}(O_A \otimes O_B)].$$

Show that this expectation factors as $\operatorname{tr}[\rho_A O_A] \operatorname{tr}[\rho_B O_B]$ for a product state. Compute the expectations of $Z \otimes I$, $I \otimes Z$, $Z \otimes Z$, and $X \otimes X$ for σ_{AB} and $|\Phi^+_{AB}\rangle$.

Solution. The operators $P_x \otimes Q_y$ are mutually orthogonal projections summing to I_{AB}. Weighting their outcome probabilities by xy gives

$$\sum_{x,y} xy \operatorname{tr}[\rho_{AB}(P_x \otimes Q_y)] = \operatorname{tr}\left[\rho_{AB}\left(\sum_{x,y} xy\, P_x \otimes Q_y\right)\right] = \operatorname{tr}[\rho_{AB}(O_A \otimes O_B)].$$

For a product state, multiplication of tensor products and the trace identity give the claimed factorization. In the two specified states, the local Z expectations vanish and the Z outcomes always agree. The operator $X \otimes X$ interchanges $|00\rangle$ and $|11\rangle$, so its diagonal entries on these vectors vanish, while it fixes $|\Phi^+_{AB}\rangle$. Hence

state	$\mathbb{E}[Z \otimes I]$	$\mathbb{E}[I \otimes Z]$	$\mathbb{E}[Z \otimes Z]$	$\mathbb{E}[X \otimes X]$
σ_{AB}	0	0	1	0
$\lvert\Phi^+_{AB}\rangle\langle\Phi^+_{AB}\rvert$	0	0	1	1

□

2.3 Purification

Preparing ρ_x with probability $p(x)$ produces the average state $\rho_A = \sum_x p(x)\rho_x$. As we saw in Chapter 1, the same density matrix can arise from different ensembles.

If we also keep the label x, we describe the preparation using a reference register X, with orthonormal basis $\{|x\rangle : x \in \Omega\}$:

$$\rho_{AX} = \sum_{x \in \Omega} p(x)\rho_x \otimes |x\rangle\langle x|.$$

This is a *classical–quantum state*: X stores classical information about the preparation of A. Its two reduced states are

$$\operatorname{tr}_X[\rho_{AX}] = \rho_A, \qquad \operatorname{tr}_A[\rho_{AX}] = \sum_x p(x)|x\rangle\langle x|.$$

Reading X in this basis returns x with probability $p(x)$. Conditional on an outcome with $p(x) > 0$, the state of A is ρ_x. For example, a subsequent POVM μ_A has conditional probabilities $p(y \mid x) = \operatorname{tr}[\mu_A(y)\rho_x]$. Discarding X loses this side information. We can also use a quantum reference system.

Definition 2.14. A *purification* of $\rho_A \in \mathcal{S}(A)$ is a unit vector $|\phi_{AR}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_R$ such that

$$\mathrm{tr}_R[|\phi_{AR}\rangle\langle\phi_{AR}|] = \rho_A.$$

We also call its density matrix a purification, and R the *reference* or *purifying system*.

Lemma 2.15. *Every state ρ_A has a purification. The purifying system can be chosen to have dimension* $\mathrm{rank}(\rho_A)$.

Proof. Let $r = \mathrm{rank}(\rho_A)$ and take a spectral decomposition on its support,

$$\rho_A = \sum_{j=0}^{r-1} p_j |e_j\rangle\langle e_j|, \qquad p_j > 0.$$

Set $\mathcal{H}_R = \mathbb{C}^r$ and $|\phi_{AR}\rangle = \sum_{j=0}^{r-1} \sqrt{p_j}|e_j\rangle \otimes |j\rangle$. This vector is normalized because $\sum_j p_j = 1$. Moreover,

$$\begin{aligned}\mathrm{tr}_R[|\phi_{AR}\rangle\langle\phi_{AR}|] &= \sum_{j,k=0}^{r-1} \sqrt{p_j p_k}|e_j\rangle\langle e_k| \,\mathrm{tr}[|j\rangle\langle k|] \\ &= \sum_{j=0}^{r-1} p_j |e_j\rangle\langle e_j| = \rho_A.\end{aligned}$$

□

The calculation only needs the reference vectors $|j\rangle$ to be orthonormal. Any pure-state ensemble $\rho_A = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ therefore gives a purification $\sum_j \sqrt{p_j}|\psi_j\rangle \otimes |j\rangle$. This construction need not use the smallest possible reference system.

Exercise 2.16. Find a purification of

$$\rho_A = \frac{1}{3}(|0\rangle\langle 0| + |+\rangle\langle +| + |1\rangle\langle 1|).$$

Solution. Using the three ensemble labels as an orthonormal basis of R, take

$$|\phi_{AR}\rangle = \frac{1}{\sqrt{3}}(|0\rangle|0\rangle + |+\rangle|1\rangle + |1\rangle|2\rangle).$$

Tracing out R removes the cross terms and gives ρ_A. □

Lemma 2.17 (Uniqueness of purification up to an isometry)**.** *If $|\phi_{AR}\rangle$ and $|\psi_{AS}\rangle$ purify the same state ρ_A and $|R| \le |S|$, then there is an isometry $V : \mathcal{H}_R \to \mathcal{H}_S$ such that*

$$(I_A \otimes V)|\phi_{AR}\rangle = |\psi_{AS}\rangle.$$

If $|R| = |S|$, this isometry is unitary.

The proof is left as homework.

2.4 Schmidt Decomposition

Recall the *singular value decomposition* (SVD) from linear algebra. For $M \in \mathrm{Lin}(\mathcal{H}, \mathcal{K})$ of rank r, there are orthonormal bases $\{|e_j\rangle\}$ of $\mathcal{K}$ and $\{|g_j\rangle\}$ of $\mathcal{H}$ and positive numbers $s_1 \geq \cdots \geq s_r > 0$ such that

$$M = \sum_{j=1}^{r} s_j |e_j\rangle\langle g_j|.$$

The s_j are the nonzero singular values. The right singular vectors $|g_j\rangle$ are eigenvectors of $M^\dagger M$, and the left singular vectors $|e_j\rangle$ are eigenvectors of $MM^\dagger$, with eigenvalues s_j^2. To pair them correctly, take $|e_j\rangle = M|g_j\rangle / s_j$ for $s_j > 0$.

We obtain a corresponding decomposition of a bipartite vector by arranging its coefficients into a matrix.

Theorem 2.18 (Schmidt decomposition)**.** *Let $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ be a pure quantum state. There are orthonormal bases $\{|e_j\rangle\}$ of $\mathcal{H}_A$ and $\{|f_j\rangle\}$ of $\mathcal{H}_B$ and positive numbers $s_1 \geq \cdots \geq s_r > 0$, where $r \leq \min(|A|, |B|)$, such that*

$$|\psi_{AB}\rangle = \sum_{j=1}^{r} s_j |e_j\rangle \otimes |f_j\rangle, \qquad \sum_{j=1}^{r} s_j^2 = 1.$$

The s_j are called Schmidt coefficients *and r is the* Schmidt rank*.*

Proof. Fix orthonormal bases $\{|a\rangle\}$ and $\{|b\rangle\}$, and let M be the $|A| \times |B|$ matrix with entries $M_{ab} = \langle ab \mid \psi_{AB}\rangle$. Choose an SVD $M = \sum_{j=1}^{r} s_j |e_j\rangle\langle g_j|$ and define

$$|f_j\rangle = \sum_b \overline{\langle b \mid g_j\rangle} |b\rangle.$$

Thus $|f_j\rangle$ is the coordinatewise conjugate of $|g_j\rangle$ in the chosen basis of B. Conjugation preserves orthonormality, and

$$M_{ab} = \sum_j s_j \langle a \mid e_j\rangle \langle g_j \mid b\rangle = \sum_j s_j \langle a \mid e_j\rangle \langle b \mid f_j\rangle.$$

Substituting these coefficients into the vector gives

$$\begin{aligned} |\psi_{AB}\rangle &= \sum_{a,b} M_{ab} |a\rangle \otimes |b\rangle \\ &= \sum_j s_j \left(\sum_a \langle a \mid e_j\rangle |a\rangle \right) \otimes \left(\sum_b \langle b \mid f_j\rangle |b\rangle \right) = \sum_j s_j |e_j\rangle \otimes |f_j\rangle. \end{aligned}$$

The product vectors in this sum are orthonormal, so normalization implies $\sum_j s_j^2 = 1$. □

Exercise 2.19. Compute the Schmidt decomposition of the state

$$|\psi_{AB}\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 1 \\ 2 \\ -2 \end{pmatrix}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Solution. Reshape the vector into its coefficient matrix and compute

$$M = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}, \qquad MM^\dagger = \begin{pmatrix} 1/5 & 0 \\ 0 & 4/5 \end{pmatrix}, \qquad M^\dagger M = \begin{pmatrix} 1/2 & -3/10 \\ -3/10 & 1/2 \end{pmatrix}.$$

In decreasing order the singular values are $s_1 = 2/\sqrt{5}$ and $s_2 = 1/\sqrt{5}$. The corresponding right singular vectors are $|g_1\rangle = |-\rangle$ and $|g_2\rangle = |+\rangle$, while $M|-\rangle/s_1 = |1\rangle$ and $M|+\rangle/s_2 = |0\rangle$. The right singular vectors are real, so conjugation leaves them unchanged. Hence

$$|\psi_{AB}\rangle = \frac{2}{\sqrt{5}}|1\rangle \otimes |-\rangle + \frac{1}{\sqrt{5}}|0\rangle \otimes |+\rangle.$$

Expanding the two terms recovers the original four entries, including the minus sign on $|11\rangle$. □

Lemma 2.20. *If $|\psi_{AB}\rangle = \sum_{j=1}^{r} s_j |e_j\rangle \otimes |f_j\rangle$ is a Schmidt decomposition, then its reduced states are*

$$\rho_A = \sum_{j=1}^{r} s_j^2 |e_j\rangle\langle e_j|, \qquad \rho_B = \sum_{j=1}^{r} s_j^2 |f_j\rangle\langle f_j|.$$

In particular, both reduced states have rank r and the same nonzero eigenvalues $s_1^2, \ldots, s_r^2$, counted with multiplicity.

Proof. First expand the density matrix of the joint pure state:

$$\rho_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}| = \sum_{j,k=1}^{r} s_j s_k |e_j\rangle\langle e_k| \otimes |f_j\rangle\langle f_k|.$$

By linearity of the partial trace and Eq. (2.2),

$$\begin{aligned} \rho_A = \operatorname{tr}_B[\rho_{AB}] &= \sum_{j,k} s_j s_k |e_j\rangle\langle e_k| \operatorname{tr}[|f_j\rangle\langle f_k|] \\ &= \sum_{j,k} s_j s_k \delta_{jk} |e_j\rangle\langle e_k| = \sum_{j=1}^{r} s_j^2 |e_j\rangle\langle e_j|. \end{aligned}$$

Here $\operatorname{tr}[|f_j\rangle\langle f_k|] = \langle f_k \mid f_j\rangle = \delta_{jk}$ because the Schmidt vectors $|f_j\rangle$ are orthonormal. Thus every term with $j \neq k$ vanishes, and the surviving terms have coefficient s_j^2. Tracing out A instead uses the orthonormality of the $|e_j\rangle$:

$$\begin{aligned} \rho_B = \operatorname{tr}_A[\rho_{AB}] &= \sum_{j,k} s_j s_k \operatorname{tr}[|e_j\rangle\langle e_k|] |f_j\rangle\langle f_k| \\ &= \sum_{j,k} s_j s_k \delta_{jk} |f_j\rangle\langle f_k| = \sum_{j=1}^{r} s_j^2 |f_j\rangle\langle f_j|. \end{aligned}$$

These are spectral decompositions: $|e_j\rangle$ is an eigenvector of ρ_A with eigenvalue s_j^2, and $|f_j\rangle$ is an eigenvector of ρ_B with the same eigenvalue. Vectors orthogonal to the respective Schmidt spans are sent to zero. Since all $s_j > 0$, both reduced states have rank r and the same nonzero eigenvalues, counted with multiplicity. □

Thus the ordered Schmidt coefficients and the Schmidt rank are uniquely determined, though the Schmidt vectors need not be. The lemma also shows that any purification of ρ_A requires $|R| \geq \operatorname{rank}(\rho_A)$. The dimension in Lemma 2.15 is therefore minimal.

Corollary 2.21. *A pure bipartite state is a product state if and only if ρ_A is pure, if and only if ρ_B is pure.*

Proof. A reduced state is pure exactly when its rank, and hence the Schmidt rank, is one. A one-term Schmidt decomposition is a product vector. Conversely, a product vector has pure reduced states. □

2.5 Entanglement

Two classical random variables are independent exactly when their joint state is a product state. Otherwise they are correlated, as in $\sigma_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Quantum systems also admit entanglement, a form of correlation that cannot be produced by shared classical randomness and local state preparation. We first define it for pure states.

Definition 2.22. A pure state $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *entangled* if it is not a product vector: there are no unit vectors $|\psi_A\rangle$ and $|\psi_B\rangle$ with $|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$.

Lemma 2.23. *A pure bipartite state is entangled if and only if either reduced state is mixed, equivalently, if and only if its Schmidt rank is greater than one.*

Proof. This is Corollary 2.21 and Lemma 2.20. □

Definition 2.24. For $|A| = |B| = d \geq 2$, a pure state on AB is *maximally entangled* if both reduced states are maximally mixed:

$$\rho_A = I_A/d, \qquad \rho_B = I_B/d.$$

We use the equal-dimension convention here. By Lemma 2.20, a pure state can have both marginals maximally mixed on their whole spaces only when $|A| = |B|$. In equal dimensions it is maximally entangled exactly when it has Schmidt rank d and every Schmidt coefficient is $1/\sqrt{d}$. Thus such states have the form

$$\frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |e_j\rangle \otimes |f_j\rangle,$$

where the two collections are orthonormal bases. In particular, using fixed standard bases, we write

$$|\Phi^+_{AB}\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j\rangle \otimes |j\rangle. \tag{2.3}$$

These states are entangled because their marginals are mixed. Later, entropy will let us quantify the connection between the mixedness of a marginal and the entanglement of a pure joint state.

Lemma 2.25 (Properties of the standard maximally entangled state)**.** *Let $|\Phi^+_{AB}\rangle$ be as in Eq. (2.3). For $d \times d$ matrices M, N expressed in those bases:*

(a) The reduced states are $\rho_A = I_A/d$ and $\rho_B = I_B/d$.

(b) $(M \otimes I)|\Phi^+_{AB}\rangle = (I \otimes M^{\mathsf{T}})|\Phi^+_{AB}\rangle$.

(c) $\langle\Phi^+_{AB}|M\otimes N|\Phi^+_{AB}\rangle = \frac{1}{d}\operatorname{tr}[M^{\mathsf{T}}N] = \frac{1}{d}\operatorname{tr}[MN^{\mathsf{T}}]$.

Proof. Part (a) follows from Lemma 2.20. For (b),

$$(M\otimes I)|\Phi^+_{AB}\rangle = \frac{1}{\sqrt{d}}\sum_{a,b} M_{ab}|a\rangle|b\rangle = (I\otimes M^{\mathsf{T}})|\Phi^+_{AB}\rangle.$$

For (c), expand in the same bases:

$$\langle\Phi^+_{AB}|M\otimes N|\Phi^+_{AB}\rangle = \frac{1}{d}\sum_{a,b} M_{ab}N_{ab} = \frac{1}{d}\operatorname{tr}[M^{\mathsf{T}}N] = \frac{1}{d}\operatorname{tr}[MN^{\mathsf{T}}].$$

□

The transpose in these identities is not the adjoint. For a maximally entangled vector defined using other paired bases $\{|e_j\rangle\}$ and $\{|f_j\rangle\}$, the same identities hold when the matrix entries are taken in those respective bases.

Exercise 2.26. Show that

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle).$$

Solution. The Hadamard matrix $H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$ satisfies $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, and $HH^{\mathsf{T}} = I$. By Lemma 2.25(b),

$$(H\otimes H)|\Phi^+_{AB}\rangle = (I\otimes HH^{\mathsf{T}})|\Phi^+_{AB}\rangle = |\Phi^+_{AB}\rangle.$$

Applying $H\otimes H$ to the standard-basis expansion proves the claim. For a general unitary U, the corresponding invariance is $(U\otimes\overline{U})|\Phi^+_{AB}\rangle = |\Phi^+_{AB}\rangle$, since $\overline{U}U^{\mathsf{T}} = I$; one cannot replace $\overline{U}$ by U in general. □

Every classical joint state $\rho_{XY} = \sum_{x,y} p_{XY}(x,y)|x\rangle\langle x|\otimes|y\rangle\langle y|$ is a convex combination of product states. This motivates the definition for arbitrary, possibly mixed, states.

Definition 2.27. A state $\rho_{AB}\in\mathcal{S}(AB)$ is *separable* if there are a finite set Ω, a probability distribution p on Ω, and states $\rho_{A,x}\in\mathcal{S}(A)$ and $\rho_{B,x}\in\mathcal{S}(B)$ such that

$$\rho_{AB} = \sum_{x\in\Omega} p(x)\rho_{A,x}\otimes\rho_{B,x}.$$

A state is *entangled* if it is not separable.

Operationally, Alice and Bob can prepare a separable state by sharing a classical random variable x, then locally preparing $\rho_{A,x}$ and $\rho_{B,x}$. Product states are separable, but separable states can be correlated: σ_{AB} is an example. States diagonal in a product basis are also separable. The converse does not hold in general, since the local states in a separable decomposition need not be diagonal in common local bases. In particular, an entangled state cannot be diagonal in any product basis.

The two definitions of entanglement agree on pure states. To see this, suppose $|\psi\rangle\langle\psi| = \sum_x p(x)\tau_x$ with $\tau_x = \rho_{A,x} \otimes \rho_{B,x}$ and $p(x) > 0$. For every $|v\rangle$ orthogonal to $|\psi\rangle$,

$$0 = \sum_x p(x)\langle v|\tau_x|v\rangle.$$

Each summand is nonnegative, so $\langle v|\tau_x|v\rangle = \|\sqrt{\tau_x}|v\rangle\|^2 = 0$ for every x. Thus every τ_x is supported on the one-dimensional span of $|\psi\rangle$ and, having trace one, equals $|\psi\rangle\langle\psi|$. Consequently this pure state is a product state. Its factors are pure, because

$$1 = \operatorname{rank}(\rho_{A,x} \otimes \rho_{B,x}) = \operatorname{rank}(\rho_{A,x}) \operatorname{rank}(\rho_{B,x}).$$

Conversely, any pure product state is separable by definition.

Overview of bipartite states. Figure 2.1 summarizes these distinctions. Here, as for classical registers earlier in the notes, *classical* means diagonal in a *fixed* product basis:

$$\rho_{AB} = \sum_{a,b} p(a,b)|a\rangle\langle a| \otimes |b\rangle\langle b|.$$

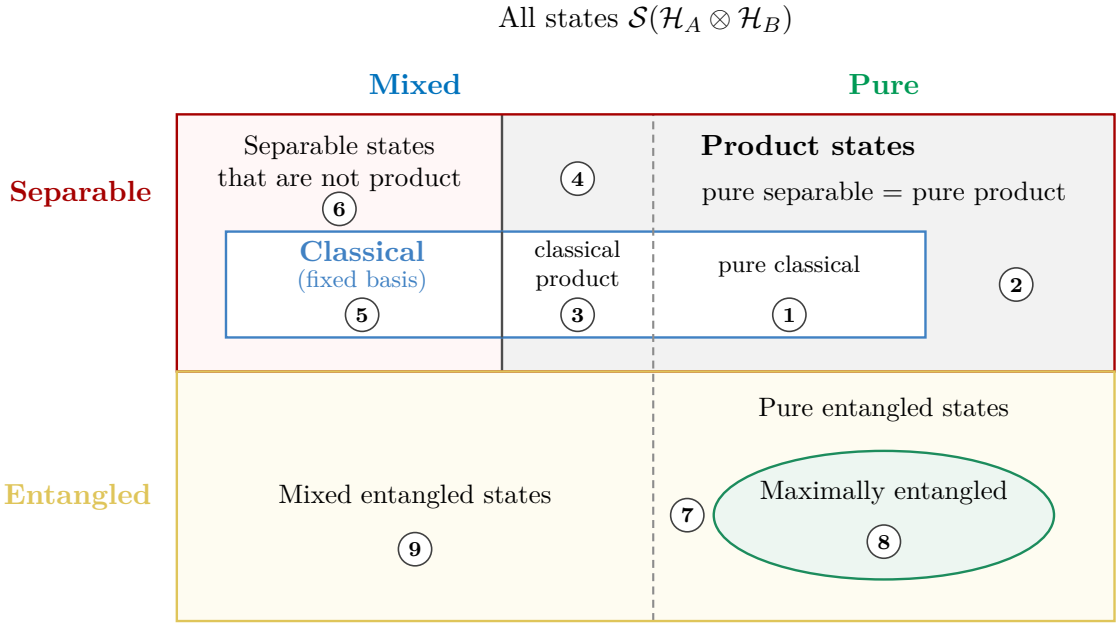
Every such classical state is separable. It is a product state exactly when $p(a,b) = p_A(a)p_B(b)$, and it is pure exactly when one probability equals one. Thus pure classical states are the projectors onto the chosen product-basis vectors, whereas pure separable states are all pure product states. Product states need not be diagonal in this fixed basis: for example, $|+\rangle\langle+| \otimes I/2$ is not diagonal in the computational basis.

If instead we allow the local bases to depend on the state, every product state is diagonal in some product basis: take eigenbases of its two factors. Under that convention, the product states form a subset of the classical states. The fixed-basis convention in the figure is therefore essential.

For entry 6 in Fig. 2.1, separability follows from the displayed mixture of product states. The vectors $|00\rangle$ and $|1+\rangle$ are orthogonal, so τ_{AB} has rank two and is mixed. Its marginals are

$$\tau_A = I_2/2, \qquad \tau_B = \frac{1}{2}(P_0 + P_+) = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Both marginals have rank two, so their product has rank four and cannot equal τ_{AB}. Thus the state is correlated. It is nonclassical in the fixed computational basis because $\langle 10|\tau_{AB}|11\rangle = 1/4$ is nonzero. The following exercise justifies entry 9 and illustrates why reduced states alone do not determine whether a mixed joint state is entangled.



Write $P_v = |v\rangle\langle v|$ and $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

No.	State	Classification
1	P_{00}	Pure, classical, product
2	P_{+0}	Pure, nonclassical, product
3	$I_4/4$	Mixed, classical, product
4	$P_+ \otimes I_2/2$	Mixed, nonclassical, product
5	$\sigma_{AB} = \frac{1}{2}(P_{00} + P_{11})$	Mixed, classical, correlated
6	$\tau_{AB} = \frac{1}{2}(P_{00} + P_{1+})$	Mixed, nonclassical, separable, correlated
7	P_ψ, $\|\psi\rangle = \frac{\sqrt{3}}{2}\|00\rangle + \frac{1}{2}\|11\rangle$	Pure entangled, not maximally entangled
8	P_{Φ^+}	Maximally entangled
9	$\frac{1}{2}\sigma_{AB} + \frac{1}{2}P_{\Phi^+}$	Mixed entangled

Figure 2.1: Schematic relationships between two-qubit state classes. The numbered points identify the examples in the table. Classical means diagonal in the fixed computational product basis. Every pure separable state is a product state. Shapes and areas do not describe the geometry or relative sizes of the sets.

Exercise 2.28 (Same reduced states, different correlations)**.** Let $\sigma_{AB} = \frac{1}{2}(P_{00} + P_{11})$, where $P_v = |v\rangle\langle v|$, and let $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. For $0 \le t \le 1$, consider

$$\rho_t = (1-t)\sigma_{AB} + tP_{\Phi^+} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & t \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ t & 0 & 0 & 1 \end{pmatrix}.$$

(a) Find the eigenvalues of ρ_t. For which t is it pure? For which t is it classical in the computational product basis?

(b) Compute both reduced states. Is ρ_t ever a product state?

(c) Prove that every separable two-qubit state ω satisfies

$$\mathrm{tr}[\omega(X \otimes X + Z \otimes Z)] \le 1.$$

Hint: First use Bloch vectors and Cauchy–Schwarz for product states, then use convexity.

(d) Evaluate this expression for ρ_t, determine exactly when it is entangled, and locate the family in Fig. 2.1.

Solution. (a) The nonzero block on the span of $|00\rangle, |11\rangle$ has eigenvectors $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$. The four eigenvalues are

$$\frac{1+t}{2}, \quad \frac{1-t}{2}, \quad 0, \quad 0.$$

Thus ρ_t is pure exactly when $t = 1$; equivalently, $\mathrm{tr}[\rho_t^2] = (1+t^2)/2$ equals one exactly then. Its only possible off-diagonal entries are $t/2$, so it is classical in the fixed basis exactly when $t = 0$.

(b) Tracing out either qubit removes the off-diagonal terms, leaving

$$(\rho_t)_A = (\rho_t)_B = I_2/2 \qquad \text{for every } t \in [0,1].$$

If ρ_t were a product state, it would equal the product of its marginals, $I_4/4$. This is impossible: its probability of 01 in the computational basis is zero, whereas that probability is $1/4$ for $I_4/4$.

(c) For a product state $\alpha \otimes \beta$, write the two Bloch vectors as $a = (a_x, a_y, a_z)$ and $b = (b_x, b_y, b_z)$, with $\|a\|, \|b\| \le 1$. By the factorization of product-state expectations,

$$\begin{aligned} \mathrm{tr}[(\alpha \otimes \beta)(X \otimes X + Z \otimes Z)] &= a_x b_x + a_z b_z \\ &\le \sqrt{a_x^2 + a_z^2}\sqrt{b_x^2 + b_z^2} \le 1. \end{aligned}$$

If $\omega = \sum_j p_j \alpha_j \otimes \beta_j$ is separable, linearity of the trace gives an average of quantities at most one. This proves the same bound for ω.

(d) The state ρ_t is supported on $|00\rangle, |11\rangle$, both of which have $Z \otimes Z$ eigenvalue $+1$. The operator $X \otimes X$ interchanges these two vectors, so

$$\mathrm{tr}[\rho_t(Z \otimes Z)] = 1, \qquad \mathrm{tr}[\rho_t(X \otimes X)] = t.$$

Their sum is $1 + t$, which exceeds the separable bound whenever $t > 0$. For $t = 0$, the state is the displayed separable mixture σ_{AB}. Therefore the complete classification is

$t = 0$	mixed, classical, separable, correlated (point 5),
$0 < t < 1$	mixed entangled (point 9 represents $t = 1/2$),
$t = 1$	pure maximally entangled (point 8).

The reduced states are the same throughout, although the joint correlations change.

The observable $W = I_{AB} - X \otimes X - Z \otimes Z$ has nonnegative expectation on every separable state and expectation $-t$ on ρ_t. Such an observable is called an *entanglement witness*: a negative expectation certifies entanglement.[1] □

[1] This witness is the two-qubit case of Eq. (4) in G. Tóth and O. Gühne, *Two measurement settings can suffice to verify multipartite entanglement*, arXiv:quant-ph/0409132. The proof here uses only Bloch vectors and convexity.

Chapter 3

Non-local Games and Quantum Foundations

Non-local games ask how well two players can coordinate their answers without communicating. Shared entanglement can improve their success probability beyond what shared randomness allows. We demonstrate this with the CHSH game, prove its optimal quantum value, and show that attaining this value identifies a Bell pair inside the shared state.

3.1 Basic Definitions

Definition 3.1. A *non-local game* is a tuple $G = (I_A, I_B, O_A, O_B, \pi, V)$, where I_A, I_B are finite nonempty sets of *questions*, O_A, O_B are finite nonempty sets of *answers*, π is a probability distribution on $I_A \times I_B$, and

$$V : O_A \times O_B \times I_A \times I_B \longrightarrow \{0, 1\}$$

is a *winning predicate*.
The referee selects (x, y) according to π, sends x to Alice and y to Bob, and receives answers $a \in O_A$ and $b \in O_B$. The players win if $V(a, b, x, y) = 1$. They can agree on a strategy and share resources before receiving their questions, but cannot communicate afterwards. The referee's questions are chosen independently of those shared resources.

Definition 3.2. A *conditional distribution* for a game G is an array $p = (p(a, b \mid x, y))_{a,b,x,y}$ of nonnegative numbers satisfying

$$\sum_{a \in O_A} \sum_{b \in O_B} p(a, b \mid x, y) = 1 \qquad \text{for every } (x, y) \in I_A \times I_B.$$

A strategy specifies such a distribution. Which distributions are allowed depends on the resources available to the players.

The *success probability* of a strategy p is

$$\omega(G, p) = \sum_{x,y} \pi(x, y) \sum_{a,b} V(a, b, x, y) p(a, b \mid x, y).$$

A strategy is *perfect* if $\omega(G,p) = 1$. For a nonempty class $\mathcal{S}$ of allowed strategies, the corresponding *value* is

$$\omega(G,\mathcal{S}) = \sup_{p\in\mathcal{S}} \omega(G,p).$$

Definition 3.3. A *deterministic strategy* is a pair of functions $f : I_A \to O_A$ and $g : I_B \to O_B$. Alice answers $a = f(x)$ and Bob answers $b = g(y)$, so

$$p(a,b \mid x,y) = \begin{cases} 1, & a = f(x) \text{ and } b = g(y), \\ 0, & \text{otherwise.} \end{cases}$$

Its winning probability is

$$\omega(G; f, g) = \sum_{x,y} \pi(x,y) V(f(x), g(y), x, y).$$

Definition 3.4. In a *classical strategy*, or *randomized strategy*, Alice and Bob share the outcome $\lambda \in \Lambda$ of a discrete random variable with probability distribution p_Λ, independently of the questions. Given λ and their questions, Alice answers a with probability $p_A(a \mid x, \lambda)$ and Bob answers b with probability $p_B(b \mid y, \lambda)$. Their local random choices are independent conditional on x, y, λ, so

$$p(a,b \mid x,y) = \sum_{\lambda\in\Lambda} p_\Lambda(\lambda) p_A(a \mid x,\lambda) p_B(b \mid y,\lambda).$$

Here $p_A(\cdot \mid x,\lambda)$ and $p_B(\cdot \mid y,\lambda)$ are probability distributions on O_A and O_B, respectively. A correlation is *classical* if it has this form.

The *classical value* is

$$\omega_c(G) = \sup_{p \text{ classical}} \omega(G,p).$$

Lemma 3.5. *Suppose a randomized strategy p wins a non-local game G with success probability $\omega(G,p)$. Then there exists a deterministic strategy for G with success probability at least $\omega(G,p)$.*

Definition 3.6. A conditional distribution p is *non-signalling* if

$$\sum_a p(a,b \mid x,y) = \sum_a p(a,b \mid x',y) \quad \text{for all } b,x,x',y,$$
$$\sum_b p(a,b \mid x,y) = \sum_b p(a,b \mid x,y') \quad \text{for all } a,x,y,y'.$$

Thus each player's marginal distribution is independent of the other player's question.

Definition 3.7. In a *finite-dimensional quantum strategy*, Alice and Bob share a state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$, chosen independently of the questions, where both local Hilbert spaces are finite-dimensional. For each $x \in I_A$, Alice has a POVM $\{\mu_A^x(a) : a \in O_A\}$ on $\mathcal{H}_A$; for each $y \in I_B$, Bob has a POVM $\{\mu_B^y(b) : b \in O_B\}$ on $\mathcal{H}_B$. They answer with their measurement outcomes, giving

$$p(a,b \mid x,y) = \operatorname{tr}\big[(\mu_A^x(a) \otimes \mu_B^y(b))\rho_{AB}\big].$$

The *quantum value* $\omega_q(G)$ is the supremum of the success probabilities over all such strategies, allowing arbitrary finite local dimensions.

The POVM conditions ensure that these probabilities are nonnegative and sum to one. They also imply that every quantum strategy is non-signalling: for example,

$$\sum_b p(a,b \mid x,y) = \operatorname{tr}\big[(\mu_A^x(a) \otimes I_B)\rho_{AB}\big],$$

which is independent of y. The other marginal is analogous. This is the no-communication calculation from Section 2.2.

Every classical strategy also has a quantum realization. Using the homework reduction to a finite mixture of deterministic strategies, store the shared label in matching orthogonal states and let each player read it before applying the corresponding deterministic response. In particular, $\omega_c(G) \le \omega_q(G)$ for every game G.

Lemma 3.8. *Suppose a quantum strategy p using a separable state ρ_{AB} wins a non-local game G with success probability $\omega(G,p)$. Then there exists a randomized, and hence also a deterministic, strategy for G with success probability at least $\omega(G,p)$.*

Thus exceeding the classical value requires entanglement.

3.2 The CHSH Game

The *Clauser–Horne–Shimony–Holt (CHSH) game* has $I_A = I_B = O_A = O_B = \{0,1\}$. The referee chooses each question pair with probability $\pi(x,y) = 1/4$. Alice and Bob win precisely when

$$a \oplus b = xy,$$

where $\oplus$ denotes addition modulo two. They must give different answers when $x = y = 1$, and the same answer for the other three question pairs.

The classical value. By Lemma 3.5, it suffices to consider $a = f(x)$ and $b = g(y)$. If a deterministic strategy won for all four question pairs, we would have

$$\begin{aligned} f(0) \oplus g(0) &= 0, \quad f(0) \oplus g(1) = 0, \\ f(1) \oplus g(0) &= 0, \quad f(1) \oplus g(1) = 1. \end{aligned}$$

Adding these equations modulo two gives $0 = 1$, because each function value occurs twice on the left. Hence every deterministic strategy loses on at least one question pair, and $\omega_c(G) \le 3/4$. Always answering $a = b = 0$ wins on exactly three pairs, so

$$\omega_c(G) = \frac{3}{4}. \tag{3.1}$$

Correlators and bias. We encode the answers $a, b \in \{0,1\}$ as the signs $(-1)^a, (-1)^b$. For an arbitrary binary POVM, define

$$A_x = \mu_A^x(0) - \mu_A^x(1), \qquad B_y = \mu_B^y(0) - \mu_B^y(1).$$

These operators are Hermitian. For projective measurements, their eigenvalues lie in $\{-1, 1\}$ and they satisfy $(A_x)^2 = I_A$ and $(B_y)^2 = I_B$. For any shared state and binary POVMs, the correlator is

$$\begin{aligned} E_{xy} := \mathrm{tr}\big[(A_x \otimes B_y)\rho_{AB}\big] &= \sum_{a,b} (-1)^{a+b} p(a, b \mid x, y) \\ &= \mathbb{P}(a = b \mid x, y) - \mathbb{P}(a \neq b \mid x, y). \end{aligned} \tag{3.2}$$

Consequently,

$$\mathbb{P}(\text{win} \mid x, y) = \frac{1 + (-1)^{xy} E_{xy}}{2}.$$

The *bias* of the strategy is

$$\begin{aligned} \beta &:= 2\mathbb{P}(\text{win}) - 1 = \mathbb{P}(\text{win}) - \mathbb{P}(\text{lose}) \\ &= \frac{1}{4}(E_{00} + E_{01} + E_{10} - E_{11}) = \frac{1}{4}\,\mathrm{tr}[M_{AB}\rho_{AB}], \end{aligned} \tag{3.3}$$

where the *CHSH operator* is

$$M_{AB} = A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1. \tag{3.4}$$

A quantum strategy. For $\theta \in \mathbb{R}$, consider the orthonormal basis

$$|\psi_0(\theta)\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \qquad |\psi_1(\theta)\rangle = -\sin\theta|0\rangle + \cos\theta|1\rangle.$$

The two basis vectors have Bloch vectors $\vec{r}_\theta = (\sin 2\theta, 0, \cos 2\theta)$ and $-\vec{r}_\theta$. The observable assigning outcomes $+1$ and -1 to them is

$$\begin{aligned} O_\theta &= |\psi_0(\theta)\rangle\langle\psi_0(\theta)| - |\psi_1(\theta)\rangle\langle\psi_1(\theta)| \\ &= \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} = \cos 2\theta\, Z + \sin 2\theta\, X. \end{aligned}$$

Alice and Bob share the Bell state

$$|\psi\rangle = |\Phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Alice uses

$$A_0 = O_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z, \qquad A_1 = O_{\pi/4} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X.$$

Thus she measures in the standard basis for $x = 0$, and in the basis $\{|+\rangle, |-\rangle\}$ for $x = 1$. Bob uses

$$B_0 = O_{\pi/8} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{Z + X}{\sqrt{2}},$$

$$B_1 = O_{-\pi/8} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} = \frac{Z - X}{\sqrt{2}}.$$

His Bloch directions are $(1, 0, 1)/\sqrt{2}$ and $(-1, 0, 1)/\sqrt{2}$. Writing $c = \cos(\pi/8)$ and $s = \sin(\pi/8)$, his ordered measurement bases are

	$b = 0$	$b = 1$
$y = 0$	$c\|0\rangle + s\|1\rangle$	$-s\|0\rangle + c\|1\rangle$
$y = 1$	$c\|0\rangle - s\|1\rangle$	$s\|0\rangle + c\|1\rangle$

For both players, the output bit is zero for the $+1$ outcome and one for the -1 outcome.

To evaluate the correlators, for unit vectors $\vec{r} = (r_x, r_y, r_z)$ and $\vec{s} = (s_x, s_y, s_z)$ define

$$O(\vec{r}) = r_x X + r_y Y + r_z Z = \begin{pmatrix} r_z & r_x - i r_y \\ r_x + i r_y & -r_z \end{pmatrix}.$$

By Lemma 2.25(c),

$$\begin{aligned} \langle \psi | O(\vec{r}) \otimes O(\vec{s}) | \psi \rangle &= \frac{1}{2} \operatorname{tr}[O(\vec{r})^{\mathsf{T}} O(\vec{s})] \\ &= r_x s_x - r_y s_y + r_z s_z. \end{aligned} \tag{3.5}$$

Here the minus sign comes from $Y^{\mathsf{T}} = -Y$; the Pauli matrices obey $\operatorname{tr}[\sigma_j \sigma_k] = 2\delta_{jk}$. For our four measurement pairs, this gives

$$E_{00} = E_{01} = E_{10} = \frac{1}{\sqrt{2}}, \qquad E_{11} = -\frac{1}{\sqrt{2}}.$$

By Eq. (3.3), the bias is $\beta = 1/\sqrt{2}$, so

$$\mathbb{P}(\text{win}) = \frac{1+\beta}{2} = \frac{1}{2} + \frac{1}{2\sqrt{2}} = \cos^2\left(\frac{\pi}{8}\right) \approx 0.8536.$$

Bell's theorem states that quantum correlations cannot always be reproduced by local hidden-variable models. We give a formulation in terms of non-local games and prove it using the CHSH game.

Theorem 3.9 (Bell's theorem)**.** *There exists a non-local game G such that*

$$\omega_c(G) < \omega_q(G).$$

Proof. Take G to be the CHSH game. We have shown that

$$\omega_c(G) = \frac{3}{4} < \frac{1}{2} + \frac{1}{2\sqrt{2}} \leq \omega_q(G). \qquad \square$$

The following Tsirelson bound shows that this quantum strategy is optimal.

Theorem 3.10 (Tsirelson bound)**.** *If G is the CHSH game, then*

$$\omega_q(G) = \frac{1}{2} + \frac{1}{2\sqrt{2}}.$$

Proof. It remains to prove the upper bound. We may assume that the local measurements are projective: in a later chapter we will prove that every finite family of POVMs can be realized by projective measurements on a larger Hilbert space, using the same isometric embedding of the state for every measurement choice. Applying this construction locally preserves all the correlations of a quantum strategy.

We may also assume that the shared state is pure. Indeed, for fixed measurements the success probability is linear in the state, so a mixed state cannot outperform every pure state in its spectral decomposition. Write the shared state as $|\psi\rangle\langle\psi|$, where $\||\psi\rangle\| = 1$. The operators A_x

and B_y are then Hermitian and satisfy $(A_x)^2 = I_A$, $(B_y)^2 = I_B$. For M_{AB} from Eq. (3.4), we have $4\beta = \langle\psi|M_{AB}|\psi\rangle$. Since M_{AB} is Hermitian, the Cauchy–Schwarz inequality gives

$$|\langle\psi|M_{AB}|\psi\rangle| \le \|M_{AB}|\psi\rangle\| \, \||\psi\rangle\| = \sqrt{\langle\psi|M_{AB}^2|\psi\rangle}.$$

Write $[R, S] = RS - SR$ for the *commutator*. Expanding the square,

$$\begin{aligned} M_{AB}^2 &= \big(A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1)\big)^2 \\ &= (A_0)^2 \otimes (B_0 + B_1)^2 + (A_1)^2 \otimes (B_0 - B_1)^2 \\ &\quad + A_0 A_1 \otimes (B_0 + B_1)(B_0 - B_1) \\ &\quad + A_1 A_0 \otimes (B_0 - B_1)(B_0 + B_1) \\ &= 4 I_A \otimes I_B - [A_0, A_1] \otimes [B_0, B_1]. \end{aligned}$$

For the last equality, use

$$\begin{aligned} (B_0 + B_1)^2 + (B_0 - B_1)^2 &= 4 I_B, \\ (B_0 + B_1)(B_0 - B_1) &= -[B_0, B_1], \\ (B_0 - B_1)(B_0 + B_1) &= [B_0, B_1]. \end{aligned}$$

Here $\|T\| = \sup_{\|v\|=1} \|Tv\|$ denotes the operator norm. For Hermitian R, S with $R^2 = S^2 = I$, we have $\|R\| = \|S\| = 1$ and therefore

$$\|[R, S]\| \le \|RS\| + \|SR\| \le 2\|R\|\|S\| = 2.$$

Using $\|R \otimes S\| = \|R\|\|S\|$, we obtain

$$\begin{aligned} |\langle\psi|[A_0, A_1] \otimes [B_0, B_1]|\psi\rangle| &\le \|[A_0, A_1] \otimes [B_0, B_1]\| \\ &= \|[A_0, A_1]\|\|[B_0, B_1]\| \le 4. \end{aligned}$$

Thus $\langle\psi|M_{AB}^2|\psi\rangle \le 8$, and

$$4\beta \le |\langle\psi|M_{AB}|\psi\rangle| \le \sqrt{8} = 2\sqrt{2}.$$

It follows that every quantum strategy satisfies

$$\mathbb{P}(\text{win}) = \frac{1 + \beta}{2} \le \frac{1}{2} + \frac{1}{2\sqrt{2}}.$$

The explicit quantum strategy above attains this bound, proving the claim. □

3.3 An Application: Self-testing a Bell Pair

A CHSH winning probability above $3/4$ certifies entanglement. Attaining the optimal probability tells us more: it identifies a Bell pair inside the shared state, even when the local dimensions and measurements are unknown. This is an example of *self-testing*. We prove the exact result using inner products, eigenvectors, and the Schmidt decomposition.[1]

[1] The proof follows the equality argument and change of basis in R. Cleve, *QIC 890: Entanglement and Nonlocal Effects*, lecture notes (April 22, 2019), Section 10.2 and Lemma 8.2. An equivalent algebraic starting point is given in I. Šupić and J. Bowles, *Self-testing of quantum systems: a review*, Quantum **4**, 337 (2020), Sections 4.2–4.3; arXiv:1904.10042.

The shared state cannot be determined uniquely. A local change of basis leaves all probabilities unchanged if the measurements are changed accordingly, and an additional shared system can be left unmeasured. The conclusion therefore allows local isometries and an arbitrary auxiliary state:

$$(V_A \otimes V_B)\rho_{AB}(V_A \otimes V_B)^\dagger = |\Phi^+\rangle\langle\Phi^+|_{A'B'} \otimes \sigma_{A''B''},$$

where V_A, V_B are local isometries, A', B' are qubits, and A'', B'' are auxiliary systems with joint state $\sigma_{A''B''}$. The isometries preserve inner products and need not be surjective. Throughout this section, tensor factors on the right of such an identity are grouped as $(A'B')(A''B'')$.

We first need a short linear algebra observation. Two operators R, T *anticommute* if $RT + TR = 0$.

Lemma 3.11 (Extracting a qubit)**.** *Let R, T be Hermitian operators on a finite-dimensional Hilbert space $\mathcal{H}$ such that $R^2 = T^2 = I$ and $RT = -TR$. There is a Hilbert space $\mathcal{K}$ and a unitary $W : \mathcal{H} \to \mathbb{C}^2 \otimes \mathcal{K}$ such that*

$$WRW^\dagger = Z \otimes I_{\mathcal{K}}, \qquad WTW^\dagger = X \otimes I_{\mathcal{K}}.$$

Proof. Since R is Hermitian and $R^2 = I$, every eigenvalue is $+1$ or -1. Choose an orthonormal basis $|e_1\rangle, \ldots, |e_m\rangle$ of its $+1$ eigenspace and put $|f_j\rangle = T|e_j\rangle$. Then

$$R|f_j\rangle = RT|e_j\rangle = -TR|e_j\rangle = -|f_j\rangle.$$

Thus $|f_j\rangle$ lies in the -1 eigenspace. The vectors $|f_j\rangle$ are orthonormal because $T^\dagger T = I$, and they span that eigenspace: T maps it back to the $+1$ eigenspace and $T^2 = I$. In particular, $\dim \mathcal{H} = 2m$.

Set $\mathcal{K} = \mathbb{C}^m$. Relabel this orthonormal basis by the unitary

$$W|e_j\rangle = |0\rangle \otimes |j\rangle, \qquad W|f_j\rangle = |1\rangle \otimes |j\rangle.$$

The operator R fixes $|e_j\rangle$ and negates $|f_j\rangle$, so it becomes $Z \otimes I$. The operator T interchanges $|e_j\rangle$ and $|f_j\rangle$, so it becomes $X \otimes I$. □

Theorem 3.12 (Exact self-testing from CHSH)**.** *Suppose a finite-dimensional quantum strategy wins the CHSH game with probability $\frac{1}{2} + \frac{1}{2\sqrt{2}}$. Then there are local isometries*

$$V_A : \mathcal{H}_A \to \mathbb{C}^2 \otimes \mathcal{K}_A, \qquad V_B : \mathcal{H}_B \to \mathbb{C}^2 \otimes \mathcal{K}_B$$

and a state σ on $\mathcal{K}_A \otimes \mathcal{K}_B$ such that

$$(V_A \otimes V_B)\rho_{AB}(V_A \otimes V_B)^\dagger = |\Phi^+\rangle\langle\Phi^+| \otimes \sigma. \tag{3.6}$$

Proof. We first prove the result for a pure shared state $|\psi\rangle_{AB}$ and projective measurements. We explain the general case at the end. The optimal winning probability gives, by Eq. (3.3),

$$\langle\psi|M_{AB}|\psi\rangle = 2\sqrt{2}.$$

To keep the formulas short, write A_x for $A_x \otimes I_B$ and B_y for $I_A \otimes B_y$. Operators belonging to different players commute. Define

$$Z_A = A_0, \qquad X_A = A_1, \qquad Z_B = \frac{B_0 + B_1}{\sqrt{2}}, \qquad X_B = \frac{B_0 - B_1}{\sqrt{2}}.$$

For the strategy already constructed in Section 3.2, these are precisely the local Pauli Z and X operators. We will show that every optimal strategy has this form on the part of the local spaces used by the state.

Step 1: the two players produce the same vectors. Since $(A_x)^2 = (B_y)^2 = I$, we have

$$Z_A^2 = X_A^2 = I, \qquad Z_B^2 + X_B^2 = 2I, \qquad M_{AB} = \sqrt{2}(Z_A Z_B + X_A X_B).$$

Expanding two squared norms now gives

$$\begin{aligned} &\|Z_A|\psi\rangle - Z_B|\psi\rangle\|^2 + \|X_A|\psi\rangle - X_B|\psi\rangle\|^2 \\ &\quad = \langle\psi|Z_A^2 + Z_B^2 + X_A^2 + X_B^2 - 2Z_A Z_B - 2X_A X_B|\psi\rangle \\ &\quad = 4 - \sqrt{2}\langle\psi|M_{AB}|\psi\rangle = 0. \end{aligned} \tag{3.7}$$

Each squared norm is nonnegative, so both are zero. Therefore

$$Z_A|\psi\rangle = Z_B|\psi\rangle, \qquad X_A|\psi\rangle = X_B|\psi\rangle. \tag{3.8}$$

Step 2: derive anticommutation relations on the state. Direct multiplication gives

$$Z_B X_B + X_B Z_B = (B_0)^2 - (B_1)^2 = 0.$$

We can transfer this relation to Alice using Eq. (3.8):

$$\begin{aligned} Z_A X_A|\psi\rangle = Z_A X_B|\psi\rangle = X_B Z_A|\psi\rangle = X_B Z_B|\psi\rangle, \\ X_A Z_A|\psi\rangle = X_A Z_B|\psi\rangle = Z_B X_A|\psi\rangle = Z_B X_B|\psi\rangle. \end{aligned}$$

Adding gives $(Z_A X_A + X_A Z_A)|\psi\rangle = 0$. Similarly,

$$\begin{aligned} Z_B^2|\psi\rangle = Z_B Z_A|\psi\rangle = Z_A Z_B|\psi\rangle = Z_A^2|\psi\rangle = |\psi\rangle, \\ X_B^2|\psi\rangle = X_B X_A|\psi\rangle = X_A X_B|\psi\rangle = X_A^2|\psi\rangle = |\psi\rangle. \end{aligned}$$

Step 3: restrict to the local Schmidt spaces. The identities in Step 2 hold on $|\psi\rangle$. To use Lemma 3.11, we restrict to the local spaces that actually occur in its Schmidt decomposition:

$$|\psi\rangle = \sum_{j=1}^{r} s_j |e_j\rangle \otimes |f_j\rangle, \qquad s_j > 0,$$

$$\mathcal{S}_A = \operatorname{span}\{|e_1\rangle, \ldots, |e_r\rangle\}, \qquad \mathcal{S}_B = \operatorname{span}\{|f_1\rangle, \ldots, |f_r\rangle\}.$$

If an operator L_A acting only on Alice satisfies $L_A|\psi\rangle = 0$, taking the partial inner product with $|f_j\rangle$ gives $s_j L_A|e_j\rangle = 0$. Hence L_A is zero on all of $\mathcal{S}_A$. The same argument applies on Bob's side.

The four operators also preserve their respective spaces. For example, the right-hand side of $Z_A|\psi\rangle = Z_B|\psi\rangle$ belongs to $\mathcal{S}_A \otimes \mathcal{H}_B$, since Z_B acts only on Bob. Taking the partial inner product with $|f_j\rangle$ shows that $s_j Z_A|e_j\rangle \in \mathcal{S}_A$, and hence $Z_A|e_j\rangle \in \mathcal{S}_A$. The other three cases follow in the same way from Eq. (3.8). Thus, on $\mathcal{S}_A$ and $\mathcal{S}_B$, both pairs are Hermitian, square to the identity, and anticommute. The lemma supplies local changes of basis V_A, V_B taking these pairs to $Z \otimes I, X \otimes I$.

Step 4: the extracted qubits form a Bell pair. Let

$$|\eta\rangle = (V_A \otimes V_B)|\psi\rangle.$$

The matching relations now say that $Z_{A'}|\eta\rangle = Z_{B'}|\eta\rangle$ and $X_{A'}|\eta\rangle = X_{B'}|\eta\rangle$. Multiplying by $Z_{A'}$ and $X_{A'}$, respectively, gives

$$(Z \otimes Z)_{A'B'}|\eta\rangle = |\eta\rangle, \qquad (X \otimes X)_{A'B'}|\eta\rangle = |\eta\rangle.$$

To see their consequence explicitly, write

$$|\eta\rangle = \sum_{i,j\in\{0,1\}} |ij\rangle_{A'B'} \otimes |\eta_{ij}\rangle_{A''B''}.$$

The first equation forces $|\eta_{01}\rangle = |\eta_{10}\rangle = 0$; the second forces $|\eta_{00}\rangle = |\eta_{11}\rangle$. Normalization therefore gives

$$|\eta\rangle = |\Phi^+\rangle_{A'B'} \otimes |\xi\rangle_{A''B''}, \qquad \||\xi\rangle\| = 1.$$

This proves the desired factorization for a pure state. To define V_A, V_B on the full original spaces, complete orthonormal bases on the unused directions, enlarging the auxiliary spaces if necessary. The resulting maps are isometries and agree with the maps already defined on $\mathcal{S}_A, \mathcal{S}_B$.

Mixed states and general measurements. For a mixed state ρ_{AB}, take a purification $|\psi\rangle_{ABE}$ and repeat the argument with all operators acting as the identity on E. The local-space argument uses the Schmidt decompositions across $A \mid BE$ and $B \mid AE$, respectively. The coefficients $|\eta_{ij}\rangle$ now include E, so the conclusion is

$$(V_A \otimes V_B \otimes I_E)|\psi\rangle_{ABE} = |\Phi^+\rangle_{A'B'} \otimes |\xi\rangle_{A''B''E}.$$

Tracing out E gives Eq. (3.6) with $\sigma = \mathrm{tr}_E|\xi\rangle\langle\xi|$. Finally, general POVMs are realized projectively on larger local spaces, as explained in Theorem 3.10. Composing the isometries introducing those local ancillas with the extraction maps proves the statement for the original strategy. □

For projective measurements, the same maps identify Alice's observables A_0, A_1 with Z, X, and Bob's B_0, B_1 with $(Z+X)/\sqrt{2}, (Z-X)/\sqrt{2}$, with identities on the auxiliary factors. These identifications hold on the local supports of the shared state; the proof places no restrictions on unused directions. They follow from Lemma 3.11 and the definitions of Z_B, X_B. The result is exact. A winning probability close to the optimum requires further estimates to obtain an approximate self-testing statement.

Chapter 4

Quantum Operations

So far we have described quantum states and the probabilities of measurement outcomes. We now ask how a state changes. A closed system evolves by a unitary operator. A subsystem can behave quite differently: interaction with another system can turn a pure state into a mixed state, and a measurement can change both the state and our information about it.

The common mathematical description of these processes is a *quantum operation*. We will describe operations by coupling to an environment, by sums of operators acting on the system, and by positivity conditions on linear maps. The equivalence of these descriptions connects the physical construction of a process with practical ways of calculating its effect.

4.1 Unitary Evolution and Measurement Updates

Axiom 5 (Evolution of a closed system)**.** The evolution of a closed quantum system between two fixed times is described by a unitary operator U on its Hilbert space:

$$|\psi\rangle \longmapsto U|\psi\rangle.$$

The operator depends on the physical evolution and the two times, but not on the initial state.

Unitarity preserves inner products, hence normalization and orthogonality. The evolution is reversible: applying $U^\dagger$ restores the initial state. For example, X exchanges $|0\rangle$ and $|1\rangle$, while Z preserves these two states up to phase and changes their relative phase in a superposition.

Continuous time. A continuous-time description is given by the Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle, \tag{4.1}$$

where the *Hamiltonian* H is Hermitian and $\hbar$ is a positive physical constant. In this discussion H is independent of time. Here H denotes the Hamiltonian, not the Hadamard operator used in Chapter 3. Its eigenvalues are the possible energies. If $H = \sum_j E_j|e_j\rangle\langle e_j|$, define

$$U(t) = e^{-itH/\hbar} = \sum_j e^{-itE_j/\hbar}|e_j\rangle\langle e_j|. \tag{4.2}$$

Differentiating this finite sum verifies that $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ solves Eq. (4.1). In particular, an energy eigenvector changes only by an overall phase. Its density operator is stationary, whereas

relative phases between different energy eigenvectors can change. The simple exponential formula assumes a fixed Hamiltonian; a time-dependent Hamiltonian requires solving the corresponding differential equation.

Example 4.1 (A qubit Hamiltonian)**.** Let $H = \hbar\omega Z/2$, where $\omega > 0$. Then

$$U(t) = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix}, \qquad U(t)|+\rangle = \frac{e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle}{\sqrt{2}}.$$

The probabilities in a computational-basis measurement remain $1/2, 1/2$. However, the probability of the outcome $+$ in an X-basis measurement is

$$|\langle +|U(t)|+\rangle|^2 = \cos^2(\omega t/2).$$

Thus time evolution can be detected through a changing relative phase, even when the computational-basis probabilities do not change.

Exercise 4.2 (Hamiltonians and unitary operators)**.** Show that $e^{-itH/\hbar}$ is unitary whenever H is Hermitian. Conversely, show that every unitary U can be written as e^{iK} for some Hermitian K. Is K unique?

Solution. In an orthonormal eigenbasis of H, the eigenvalues of $e^{-itH/\hbar}$ have modulus one. Explicitly,

$$U(t)^\dagger U(t) = \sum_j e^{itE_j/\hbar} e^{-itE_j/\hbar} |e_j\rangle\langle e_j| = I.$$

A unitary is a normal matrix, so the spectral theorem for normal matrices gives an orthonormal eigenbasis $|u_j\rangle$ with eigenvalues $e^{i\theta_j}$, where $\theta_j \in \mathbb{R}$. Set

$$K = \sum_j \theta_j |u_j\rangle\langle u_j|.$$

Then K is Hermitian and $e^{iK} = U$. Replacing any θ_j by $\theta_j + 2\pi n_j$, with $n_j \in \mathbb{Z}$, gives another possible K. Thus a single unitary does not determine a unique Hamiltonian. □

Evolution of density operators. Suppose $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$. Applying the same unitary to every member of the ensemble gives

$$\rho \longmapsto \sum_j p_j U|\psi_j\rangle\langle\psi_j|U^\dagger = U\rho U^\dagger. \tag{4.3}$$

The result depends only on ρ, not on its ensemble decomposition. Unitary conjugation preserves all eigenvalues of ρ, so it preserves purity as well as trace and positivity. Differentiating $\rho(t) = U(t)\rho(0)U(t)^\dagger$ gives

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}(H\rho - \rho H) = -\frac{i}{\hbar}[H, \rho]. \tag{4.4}$$

This is the *von Neumann equation*. In particular, any state commuting with H is stationary.

Measurements with an output state. In Section 1.4, a POVM specified the probabilities of the outcomes. To describe the state afterwards, we need more information. We first consider measurements with one operator per recorded outcome. A general measurement can require several operators for one outcome, and can then leave a mixed conditional state even for a pure input. We treat this more general case using quantum instruments in Section 4.5.

Axiom 6 (Measurement operators)**.** Let $M_x \colon \mathcal{H} \to \mathcal{K}$ be linear operators indexed by a finite outcome set Ω, with

$$\sum_x M_x^\dagger M_x = I_{\mathcal{H}}.$$

These operators describe a measurement with outcome probabilities and conditional output states

$$p(x) = \operatorname{tr}[M_x \rho M_x^\dagger], \qquad \rho_x = \frac{M_x \rho M_x^\dagger}{p(x)} \quad \text{if } p(x) > 0. \tag{4.5}$$

For a pure input, the conditional vector is $M_x|\psi\rangle / \|M_x|\psi\rangle\|$ whenever its denominator is nonzero. For an ensemble $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, the probability of x is the sum of the joint probabilities $p_j \|M_x|\psi_j\rangle\|^2$. Conditional on x, the weight of the jth preparation is $p_j \|M_x|\psi_j\rangle\|^2 / p(x)$. Multiplying these conditional weights by the corresponding normalized pure states yields Eq. (4.5).

The associated POVM is

$$\mu(x) = M_x^\dagger M_x, \qquad p(x) = \operatorname{tr}[\mu(x)\rho].$$

Conversely, a POVM always has at least one such realization: take $M_x = \sqrt{\mu(x)}$. It generally has many others. For projective measurements, the choice $M_x = P_x$ gives the *Lüders update* $P_x \rho P_x / \operatorname{tr}[P_x \rho]$.

Example 4.3 (The same POVM, different output states)**.** Consider the computational-basis POVM on a qubit. Both

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|, \qquad N_0 = |0\rangle\langle 0|, \quad N_1 = |0\rangle\langle 1|$$

realize it, since $M_x^\dagger M_x = N_x^\dagger N_x = |x\rangle\langle x|$. The first realization leaves the state $|x\rangle$ after outcome x. The second leaves $|0\rangle$ after either outcome. The outcome statistics therefore do not determine the residual quantum state.

If we perform a measurement but discard its outcome, the output is

$$\rho' = \sum_x p(x)\rho_x = \sum_x M_x \rho M_x^\dagger. \tag{4.6}$$

Zero-probability outcomes contribute zero to the final sum. This map is linear, while conditioning on a particular outcome generally introduces a nonlinear normalization.

Example 4.4 (An unread computational-basis measurement)**.** For $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$,

$$\begin{pmatrix} a & b \\ \bar{b} & 1-a \end{pmatrix} \longmapsto P_0 \rho P_0 + P_1 \rho P_1 = \begin{pmatrix} a & 0 \\ 0 & 1-a \end{pmatrix}.$$

The diagonal probabilities survive but the off-diagonal entries vanish. For instance, $|+\rangle\langle +|$ becomes $I/2$. This evolution cannot be unitary conjugation, since it changes a pure state

into a mixed state.

Exercise 4.5 (Successive measurements)**.** A measurement with operators L_l is followed by one with operators M_m. Show that the entire procedure, with both outcomes recorded, is a single measurement with operators $N_{lm} = M_m L_l$. Verify both the joint probabilities and the conditional output states.

Solution. The completeness condition is

$$\sum_{l,m} N_{lm}^\dagger N_{lm} = \sum_l L_l^\dagger \left(\sum_m M_m^\dagger M_m \right) L_l = \sum_l L_l^\dagger L_l = I.$$

When $p(l) > 0$, multiplying $p(l)$ by the conditional probability of m cancels the first normalization, giving

$$p(l,m) = \operatorname{tr}[M_m L_l \rho L_l^\dagger M_m^\dagger] = \operatorname{tr}[N_{lm} \rho N_{lm}^\dagger].$$

If $p(l) = 0$, the joint probability is zero as well. For $p(l,m) > 0$, the two successive normalized updates similarly reduce to

$$\rho_{lm} = \frac{N_{lm} \rho N_{lm}^\dagger}{p(l,m)}.$$

Thus the single measurement reproduces the whole procedure. □

4.2 Environments and Operator-Sum Representations

A classical comparison. A classical channel is specified by conditional probabilities $T(y \mid x)$. An input distribution p produces the output distribution

$$q(y) = \sum_x T(y \mid x) p(x), \qquad T(y \mid x) \geq 0, \quad \sum_y T(y \mid x) = 1.$$

For example, a bit flipped with probability p has transition matrix

$$\begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}.$$

The normalization of each column guarantees normalization of every output distribution. We seek an analogous description for quantum states, including their off-diagonal entries and their correlations with other systems.

Coupling to an environment. Let A be our system and E an environment. Assume their initial state is $\rho_A \otimes \sigma_E$, where σ_E is fixed independently of ρ_A. Suppose the joint system evolves by a unitary U. If we keep only A, its final state is

$$\mathcal{E}(\rho) = \operatorname{tr}_E\big[U(\rho \otimes \sigma_E)U^\dagger\big]. \tag{4.7}$$

This combines three familiar procedures: adjoining a system, unitary evolution, and discarding a system. Even though the joint evolution is reversible, the reduced evolution need not be.

We may take the environment to start in a pure state $|e_0\rangle$. Indeed, if σ_E is mixed, purify it using an additional system F, apply $U \otimes I_F$, and discard both E and F. The output on A is unchanged. Thus enlarging the environment removes the need for a mixed initial environment state.

Proposition 4.6 (From an environment to Kraus operators)**.** *Let $|e_k\rangle$ be an orthonormal basis of the environment and set*

$$K_k = (I_A \otimes \langle e_k|)U(I_A \otimes |e_0\rangle).$$

Then the map in Eq. (4.7), with $\sigma_E = |e_0\rangle\langle e_0|$, has the form

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I_A. \tag{4.8}$$

Proof. The partial-trace formula from Section 2.2 gives

$$\begin{aligned}\mathcal{E}(\rho) &= \sum_k (I_A \otimes \langle e_k|)U(\rho \otimes |e_0\rangle\langle e_0|)U^\dagger(I_A \otimes |e_k\rangle) \\ &= \sum_k K_k \rho K_k^\dagger.\end{aligned}$$

For completeness, insert the identity on the environment:

$$\begin{aligned}\sum_k K_k^\dagger K_k &= (I_A \otimes \langle e_0|)U^\dagger \left(I_A \otimes \sum_k |e_k\rangle\langle e_k| \right) U(I_A \otimes |e_0\rangle) \\ &= (I_A \otimes \langle e_0|)(I_A \otimes |e_0\rangle) = I_A.\end{aligned}$$

□

Definition 4.7 (Operator-sum representation)**.** An expression

$$\mathcal{E}(A) = \sum_k K_k A K_k^\dagger, \qquad K_k \in \mathrm{Lin}(\mathcal{H}_A, \mathcal{H}_B),$$

is called an *operator-sum representation* or *Kraus representation* of the linear map $\mathcal{E}\colon \mathrm{Lin}(\mathcal{H}_A) \to \mathrm{Lin}(\mathcal{H}_B)$. The K_k are its *Kraus operators*.

We use A for an arbitrary operator and ρ for a density operator. The input and output Hilbert spaces may differ. For example, discarding one part of a joint system reduces the dimension. A map of the form Eq. (4.8) is called a *quantum channel*; an intrinsic definition will follow in Section 4.4.

Proposition 4.8 (Trace conditions)**.** *A map with Kraus operators K_k preserves trace if and only if $\sum_k K_k^\dagger K_k = I_A$. It is trace-nonincreasing on positive operators if and only if $\sum_k K_k^\dagger K_k \le I_A$.*

Proof. Put $S = \sum_k K_k^\dagger K_k$. Cyclicity of the trace gives

$$\mathrm{tr}[\mathcal{E}(A)] = \mathrm{tr}[SA].$$

If $S = I_A$, this is $\mathrm{tr}[A]$. Conversely, equality on every pure state implies $\langle\psi|(S - I_A)|\psi\rangle = 0$ for every vector, so the Hermitian operator $S - I_A$ is zero. Likewise, $S \le I_A$ implies $\mathrm{tr}[(I_A - S)A] \ge 0$ for every positive A. The converse follows by testing $A = |\psi\rangle\langle\psi|$. □

Example 4.9 (Dephasing through an environment)**.** Let A and E be qubits, with E initially in $|0\rangle$, and let

$$U = P_0 \otimes I + P_1 \otimes X, \qquad P_j = |j\rangle\langle j|.$$

This is the controlled-NOT unitary: it flips E precisely when A is in $|1\rangle$. The Kraus operators are

$$K_0 = (I \otimes \langle 0|)U(I \otimes |0\rangle) = P_0, \qquad K_1 = (I \otimes \langle 1|)U(I \otimes |0\rangle) = P_1.$$

Discarding the environment gives the same map as the unread measurement in Example 4.4. For a pure input the joint evolution is

$$(a|0\rangle + b|1\rangle)|0\rangle \longmapsto a|00\rangle + b|11\rangle.$$

The reduced state loses its off-diagonal entries because the environment now distinguishes the two basis components. The joint state retains the coherence between $|00\rangle$ and $|11\rangle$.

If the environment in Proposition 4.6 is measured in the basis $|e_k\rangle$, outcome k has probability $p(k) = \operatorname{tr}[K_k \rho K_k^\dagger]$ and leaves the conditional state $K_k \rho K_k^\dagger / p(k)$ on A. Discarding that outcome gives $\sum_k K_k \rho K_k^\dagger$ again. Thus an individual term can be interpreted as a measurement branch. This interpretation depends on which environment measurement we choose; a channel does not specify a unique list of underlying events.

Exercise 4.10 (A different interaction)**.** Two qubits interact through

$$U = \frac{X \otimes I + Y \otimes X}{\sqrt{2}}.$$

The second qubit starts in $|0\rangle$ and is then discarded. Verify that U is unitary and find Kraus operators for the resulting channel on the first qubit.

Solution. Both summands are Hermitian. Using $XY + YX = 0$ gives

$$U^\dagger U = U^2 = \tfrac{1}{2}(I \otimes I + I \otimes I + (XY + YX) \otimes X) = I.$$

Taking matrix elements on the second qubit yields $K_0 = X/\sqrt{2}$ and $K_1 = Y/\sqrt{2}$. Hence

$$\mathcal{E}(\rho) = \tfrac{1}{2} X \rho X + \tfrac{1}{2} Y \rho Y.$$

In particular, on a general input $\rho = \begin{pmatrix} a & b \\ \bar{b} & 1-a \end{pmatrix}$ the output is $\begin{pmatrix} 1-a & 0 \\ 0 & a \end{pmatrix}$. The two unitary branches cancel the off-diagonal entries and exchange the diagonal entries. □

Exercise 4.11 (Composition of operations)**.** Let $\mathcal{E}\colon \operatorname{Lin}(\mathcal{H}_A) \to \operatorname{Lin}(\mathcal{H}_B)$ have Kraus operators K_i, and let $\mathcal{F}\colon \operatorname{Lin}(\mathcal{H}_B) \to \operatorname{Lin}(\mathcal{H}_C)$ have Kraus operators L_j. Find a Kraus representation of $\mathcal{F} \circ \mathcal{E}$. Show that composition preserves the trace-preserving and trace-nonincreasing conditions.

Solution. Direct substitution gives

$$(\mathcal{F} \circ \mathcal{E})(A) = \sum_{j,i} (L_j K_i) A (L_j K_i)^\dagger.$$

If both maps preserve trace, then

$$\sum_{i,j}(L_jK_i)^\dagger(L_jK_i) = \sum_i K_i^\dagger \left(\sum_j L_j^\dagger L_j\right) K_i = \sum_i K_i^\dagger K_i = I_A.$$

If each completeness equality is replaced by an inequality, the same calculation gives

$$\sum_{i,j}(L_jK_i)^\dagger(L_jK_i) \le \sum_i K_i^\dagger K_i \le I_A.$$

All products have the required domain and codomain even when the three spaces have different dimensions. □

4.3 Physical Realization of Operations

We have obtained Kraus operators from an environment model. The converse shows that the completeness condition is also sufficient to construct such a model.

Theorem 4.12 (Isometric realization of a channel)**.** *Suppose $K_k\colon \mathcal{H}_A \to \mathcal{H}_B$ satisfy $\sum_k K_k^\dagger K_k = I_A$. Let $\mathcal{H}_E$ have orthonormal basis $|k\rangle$, indexed by the Kraus operators. The map*

$$V\colon \mathcal{H}_A \longrightarrow \mathcal{H}_B \otimes \mathcal{H}_E, \qquad V|\psi\rangle = \sum_k K_k|\psi\rangle \otimes |k\rangle \tag{4.9}$$

is an isometry, and

$$\mathcal{E}(A) = \sum_k K_k A K_k^\dagger = \mathrm{tr}_E[VAV^\dagger].$$

Proof. For any vectors $|\psi\rangle, |\phi\rangle$,

$$\langle V\psi \mid V\phi\rangle = \sum_{k,l}\langle\psi|K_k^\dagger K_l|\phi\rangle\langle k \mid l\rangle = \langle\psi|\left(\sum_k K_k^\dagger K_k\right)|\phi\rangle = \langle\psi \mid \phi\rangle.$$

Thus $V^\dagger V = I_A$. Moreover,

$$VAV^\dagger = \sum_{k,l} K_k A K_l^\dagger \otimes |k\rangle\langle l|.$$

Since $\mathrm{tr}[|k\rangle\langle l|] = \delta_{kl}$, taking the partial trace gives the desired operator sum. □

This is the finite-dimensional *Stinespring representation*. For equal input and output spaces, it has a particularly direct unitary implementation. Choose a fixed unit vector $|e_0\rangle \in \mathcal{H}_E$ and prescribe

$$U(|\psi\rangle \otimes |e_0\rangle) = V|\psi\rangle.$$

The prescribed map is an isometry on the subspace $\mathcal{H}_A \otimes \mathrm{span}\{|e_0\rangle\}$. Extend an orthonormal basis of this subspace and its image to orthonormal bases of $\mathcal{H}_A \otimes \mathcal{H}_E$, and map the remaining basis vectors to one another. This extends the prescription to a unitary U on the whole space. Consequently,

$$\mathcal{E}(\rho) = \mathrm{tr}_E[U(\rho \otimes |e_0\rangle\langle e_0|)U^\dagger].$$

The additional columns of U are not fixed by the channel.

Different input and output spaces. An isometry $V\colon \mathcal{H}_A \to \mathcal{H}_B \otimes \mathcal{H}_E$ still has a unitary implementation with additional registers. Prepare B, E in fixed pure states and prescribe, on the input subspace of $A \otimes B \otimes E$,

$$|\psi\rangle_A|0\rangle_B|0\rangle_E \longmapsto |0\rangle_A \otimes V|\psi\rangle_A.$$

This preserves inner products and extends to a unitary on the entire space by the same basis-completion argument. Discarding A, E leaves the required output on B. Thus allowing a different output dimension does not introduce a new physical operation.

Example 4.13 (Realizing a random choice of unitaries)**.** Suppose U_k are unitary operators and p_k are probabilities. The map

$$\mathcal{E}(\rho) = \sum_k p_k U_k \rho U_k^\dagger$$

has Kraus operators $K_k = \sqrt{p_k}U_k$. Its isometry is $V|\psi\rangle = \sum_k \sqrt{p_k}U_k|\psi\rangle \otimes |k\rangle$. It can also be implemented by preparing the environment in $\sum_k \sqrt{p_k}|k\rangle$, applying the controlled unitary $\sum_k U_k \otimes |k\rangle\langle k|$, and discarding the environment. Classical random choices of unitary evolutions are therefore included in the environment model.

Exercise 4.14 (Realizing an operation with a success outcome)**.** Suppose $K_k\colon \mathcal{H} \to \mathcal{H}$ satisfy $S = \sum_k K_k^\dagger K_k \le I$. Explain how to realize $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$ by a unitary interaction with an environment followed by a measurement on the environment. Identify the success probability and the conditional output state.

Solution. Add one more operator $K_f = \sqrt{I - S}$. Then

$$\sum_k K_k^\dagger K_k + K_f^\dagger K_f = I.$$

The enlarged family defines a channel, realized by

$$V|\psi\rangle = \sum_k K_k|\psi\rangle \otimes |k\rangle + K_f|\psi\rangle \otimes |f\rangle,$$

and hence by a unitary with a fixed initial environment state. Measure the environment with the two projections $P_s = \sum_k |k\rangle\langle k|$ and $P_f = |f\rangle\langle f|$. On success, the unnormalized reduced state is exactly $\mathcal{E}(\rho)$. Its trace is

$$p(s) = \operatorname{tr}[S\rho] \le 1,$$

and, if this number is positive, the conditional state is $\mathcal{E}(\rho)/p(s)$. The failure branch has probability $\operatorname{tr}[(I - S)\rho]$. Thus loss of trace records a possible outcome, rather than loss of total probability from the full experiment. □

4.4 Complete Positivity and the Choi Operator

Can we recognize a possible quantum process without specifying its environment? First, a deterministic process should respect mixtures: if we choose ρ_j with probability p_j and apply the same process regardless of j, the output should be $\sum_j p_j \mathcal{E}(\rho_j)$.

Mixtures and a selected outcome. The same principle applies when the process describes just one possible measurement outcome. Suppose the input is $\rho = \sum_j p_j \rho_j$. On input ρ_j, let the selected outcome occur with probability q_j, leaving a conditional state σ_j when $q_j > 0$. Encode both pieces of information in the unnormalized operator $\mathcal{E}(\rho_j) = q_j \sigma_j$; set it to zero when $q_j = 0$. The total probability of the outcome is $q = \sum_j p_j q_j$. If $q > 0$, Bayes' rule gives the conditional preparation probabilities

$$\Pr(j \mid \text{outcome}) = \frac{p_j q_j}{q}.$$

Thus the conditional output state is $\sigma = \sum_{j:q_j>0} (p_j q_j / q) \sigma_j$, and

$$\mathcal{E}\left(\sum_j p_j \rho_j\right) = q\sigma = \sum_{j:q_j>0} p_j q_j \sigma_j = \sum_j p_j \mathcal{E}(\rho_j).$$

If $q = 0$, every term with $p_j > 0$ has $q_j = 0$, so both sides are zero. This motivates an *affine* map on density operators even for a selected outcome. Its trace is the outcome probability. The normalized update $\mathcal{E}(\rho)/\operatorname{tr}[\mathcal{E}(\rho)]$ generally fails to be affine because conditioning changes the preparation weights.

From states to all operators. An affine map $\mathcal{E}$ on density operators has a unique complex-linear extension to the entire operator space. We verify the construction. For a positive operator A, define

$$\widetilde{\mathcal{E}}(A) = \begin{cases} \operatorname{tr}[A]\, \mathcal{E}(A/\operatorname{tr}[A]), & A \neq 0, \\ 0, & A = 0. \end{cases}$$

A nonzero positive operator has positive trace, so this is well-defined. It is homogeneous for nonnegative scalars. If $A, B \geq 0$ have positive traces a, b, affineness gives

$$\begin{aligned} \widetilde{\mathcal{E}}(A+B) &= (a+b)\mathcal{E}\left(\frac{a}{a+b}\frac{A}{a} + \frac{b}{a+b}\frac{B}{b}\right) \\ &= \widetilde{\mathcal{E}}(A) + \widetilde{\mathcal{E}}(B). \end{aligned}$$

The same identity holds when either operator is zero.

Every Hermitian operator is a difference $H = A - B$ of positive operators, by separating its positive and negative eigenvalues. Set $\widetilde{\mathcal{E}}(H) = \widetilde{\mathcal{E}}(A) - \widetilde{\mathcal{E}}(B)$. If also $H = C - D$ with $C, D \geq 0$, then $A + D = C + B$. Additivity on positive operators therefore gives

$$\widetilde{\mathcal{E}}(A) - \widetilde{\mathcal{E}}(B) = \widetilde{\mathcal{E}}(C) - \widetilde{\mathcal{E}}(D),$$

so the definition is independent of the decomposition. Additivity and homogeneity, together with $\widetilde{\mathcal{E}}(-H) = -\widetilde{\mathcal{E}}(H)$, make this extension real-linear on Hermitian operators.

Finally, every operator T has the unique decomposition

$$T = H + iG, \qquad H = \frac{T + T^\dagger}{2}, \quad G = \frac{T - T^\dagger}{2i},$$

with H, G Hermitian. Define $\widetilde{\mathcal{E}}(T) = \widetilde{\mathcal{E}}(H) + i\widetilde{\mathcal{E}}(G)$. Real-linearity on the Hermitian part makes this map complex-linear. Every step is forced by linearity and the original values on density operators, proving uniqueness. We henceforth write $\mathcal{E}$ for this extension and define linear maps on the entire operator space.

Definition 4.15 (Positive and completely positive maps)**.** A linear map $\mathcal{E}\colon \mathrm{Lin}(\mathcal{H}_A) \to \mathrm{Lin}(\mathcal{H}_B)$ is *positive* if $\mathcal{E}(A) \geq 0$ whenever $A \geq 0$. It is *completely positive* if $\mathrm{id}_R \otimes \mathcal{E}$ is positive for every finite-dimensional reference system R.
A *quantum operation* is a completely positive map that is trace-nonincreasing on positive operators. A *quantum channel* is a completely positive, trace-preserving map.

Complete positivity expresses consistency with an additional system on which nothing is done. Even if we act only on A, the input may be entangled with a reference R. The resulting joint operator must still be positive. We often abbreviate "completely positive and trace-preserving" as *CPTP*.

Example 4.16 (Positive is not enough)**.** Transposition $T(A) = A^T$ in a fixed basis is linear, positive, and trace-preserving. Positivity follows, for example, by transposing a spectral decomposition: if $A = \sum_j \lambda_j |v_j\rangle\langle v_j|$ with $\lambda_j \geq 0$, then $A^T = \sum_j \lambda_j |\overline{v_j}\rangle\langle\overline{v_j}| \geq 0$. Nevertheless, for $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$,

$$(\mathrm{id} \otimes T)(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The vector $(|01\rangle - |10\rangle)/\sqrt{2}$ is an eigenvector with eigenvalue $-1/2$. The other three eigenvalues are $1/2$. Thus transposition is not completely positive and cannot be a channel acting on an arbitrary subsystem.

The definition of complete positivity appears to require checking infinitely many reference systems. The following construction reduces it to positivity of one matrix.

Definition 4.17 (Choi operator)**.** Fix a basis $|i\rangle$ of $\mathcal{H}_A$ and a copy $\mathcal{H}_R$ of this space. Let $|\Omega\rangle = \sum_i |i\rangle_R \otimes |i\rangle_A$ be the *unnormalized* maximally entangled vector. The Choi operator of $\mathcal{E}\colon \mathrm{Lin}(\mathcal{H}_A) \to \mathrm{Lin}(\mathcal{H}_B)$ is

$$J_{\mathcal{E}} = (\mathrm{id}_R \otimes \mathcal{E})(|\Omega\rangle\langle\Omega|) = \sum_{i,j} |i\rangle\langle j|_R \otimes \mathcal{E}(|i\rangle\langle j|_A). \tag{4.10}$$

The tensor factors are ordered as $R \otimes B$: input reference first, output second.

The (i,j) block of $J_{\mathcal{E}}$ is $\mathcal{E}(|i\rangle\langle j|)$, so it determines the map on a basis of the operator space. More explicitly,

$$\mathcal{E}(A) = \mathrm{tr}_R[(A^T \otimes I_B) J_{\mathcal{E}}]. \tag{4.11}$$

Here A^T acts on the reference copy in the chosen basis. To verify the formula, use $\mathrm{tr}[A^T |i\rangle\langle j|] = A_{ij}$ and $A = \sum_{i,j} A_{ij} |i\rangle\langle j|$. Both the transpose and the tensor order in this formula depend on the convention in Eq. (4.10).

Theorem 4.18 (Characterizations of complete positivity)**.** *For a linear map* $\mathcal{E}\colon \mathrm{Lin}(\mathcal{H}_A) \to \mathrm{Lin}(\mathcal{H}_B)$*, the following are equivalent:*

(a) $\mathcal{E}$ *is completely positive.*

(b) $J_{\mathcal{E}} \geq 0$*.*

(c) $\mathcal{E}$ *has a Kraus representation.*

If these conditions hold, then $\mathcal{E}$ *is a channel precisely when* $\operatorname{tr}_B J_{\mathcal{E}} = I_R$*, and is a quantum operation precisely when* $\operatorname{tr}_B J_{\mathcal{E}} \leq I_R$*.*

Proof. (a) implies (b). The operator $|\Omega\rangle\langle\Omega|$ is positive. Complete positivity therefore gives $J_{\mathcal{E}} \geq 0$.

(b) implies (c). By the spectral theorem, write

$$J_{\mathcal{E}} = \sum_{k=1}^{r} |v_k\rangle\langle v_k|,$$

where each $|v_k\rangle$ includes the square root of its positive eigenvalue. Expand it in the reference basis:

$$|v_k\rangle = \sum_i |i\rangle_R \otimes |w_{ki}\rangle_B, \qquad K_k|i\rangle_A = |w_{ki}\rangle_B.$$

This specifies a linear operator $K_k\colon \mathcal{H}_A \to \mathcal{H}_B$. Comparing the (i,j) blocks of $J_{\mathcal{E}}$ gives

$$\mathcal{E}(|i\rangle\langle j|) = \sum_k |w_{ki}\rangle\langle w_{kj}| = \sum_k K_k |i\rangle\langle j| K_k^\dagger.$$

The matrix units span all operators, so linearity gives the Kraus representation for every A.

(c) implies (a). For any reference system and any positive W on $R \otimes A$,

$$(\mathrm{id}_R \otimes \mathcal{E})(W) = \sum_k (I_R \otimes K_k) W (I_R \otimes K_k)^\dagger \geq 0.$$

Each term is positive by the positivity of W.

Finally, writing $S = \sum_k K_k^\dagger K_k$, we obtain

$$\begin{aligned} \operatorname{tr}_B J_{\mathcal{E}} &= \sum_{i,j} |i\rangle\langle j| \operatorname{tr}[\mathcal{E}(|i\rangle\langle j|)] \\ &= \sum_{i,j} |i\rangle\langle j| \langle j|S|i\rangle = S^T. \end{aligned}$$

The trace conditions now follow from Proposition 4.8; transposition preserves positivity and fixes the identity. □

The proof also gives an algorithm: diagonalize the Choi operator and reshape each eigenvector, multiplied by the square root of its eigenvalue, into a Kraus operator. For a channel with input dimension d_A, $J_{\mathcal{E}}/d_A$ is a density operator. It is the output obtained by applying the channel to one half of a normalized maximally entangled state. Its reference marginal is I_R/d_A.

Corollary 4.19 (Number of Kraus operators)**.** *The minimum number of nonzero Kraus operators for a completely positive map is* $\operatorname{rank} J_{\mathcal{E}}$*. In particular, at most* $d_A d_B$ *are needed.*

Proof. The spectral construction uses $\operatorname{rank} J_{\mathcal{E}}$ terms. Conversely, every Kraus representation gives

$$J_{\mathcal{E}} = \sum_k |v_k\rangle\langle v_k|, \qquad |v_k\rangle = (I_R \otimes K_k)|\Omega\rangle.$$

The rank of this sum is at most its number of nonzero terms. □

Example 4.20 (Three descriptions of dephasing)**.** For $\mathcal{D}(\rho) = P_0\rho P_0 + P_1\rho P_1$, the Choi operator is

$$J_{\mathcal{D}} = |00\rangle\langle 00| + |11\rangle\langle 11| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

It is positive and its output partial trace is I_R. Its two nonzero eigenvectors recover the Kraus operators P_0, P_1. The corresponding isometry is

$$V|0\rangle = |00\rangle, \qquad V|1\rangle = |11\rangle,$$

which is precisely the restriction of the controlled-NOT construction in Example 4.9. The Choi rank is two, so two Kraus operators suffice and one cannot suffice.

Nonuniqueness of Kraus operators. The same dephasing channel also has Kraus operators $I/\sqrt{2}$ and $Z/\sqrt{2}$, since

$$\tfrac{1}{2}\rho + \tfrac{1}{2}Z\rho Z = P_0\rho P_0 + P_1\rho P_1.$$

Thus applying a random phase flip and performing an unread measurement produce exactly the same channel. The two lists are related by

$$P_0 = \frac{1}{\sqrt{2}}\left(\frac{I}{\sqrt{2}} + \frac{Z}{\sqrt{2}}\right), \qquad P_1 = \frac{1}{\sqrt{2}}\left(\frac{I}{\sqrt{2}} - \frac{Z}{\sqrt{2}}\right).$$

More generally, a change of orthonormal basis in the environment mixes the Kraus operators by a unitary matrix. The following theorem proves that such mixing leaves the map unchanged and accounts for all its Kraus representations.

Theorem 4.21 (Freedom in a Kraus representation)**.** *Let* $(K_1, \dots, K_m)$ *and* $(L_1, \dots, L_n)$ *be two lists of Kraus operators. Append zero operators to the shorter list so that both have length* N*. They describe the same map if and only if there is an* $N \times N$ *unitary matrix* u *such that*

$$L_j = \sum_{k=1}^{N} u_{jk} K_k \qquad (1 \le j \le N).$$

Proof. First suppose $L_j = \sum_k u_{jk} K_k$ with u unitary. Then, for every operator A,

$$\sum_j L_j A L_j^\dagger = \sum_{k,l} \left(\sum_j u_{jk}\overline{u_{jl}}\right) K_k A K_l^\dagger = \sum_k K_k A K_k^\dagger.$$

Thus unitary mixing does not change the map.

Conversely, suppose the two lists describe the same map. As in the Choi construction, associate to them the vectors

$$|v_k\rangle = (I_R \otimes K_k)|\Omega\rangle, \qquad |w_j\rangle = (I_R \otimes L_j)|\Omega\rangle.$$

Their common Choi operator satisfies

$$J = \sum_{k=1}^{N} |v_k\rangle\langle v_k| = \sum_{j=1}^{N} |w_j\rangle\langle w_j|.$$

If $J = 0$, taking traces shows that every v_k, w_j is zero, hence every Kraus operator is zero; any unitary u works. Otherwise, take the spectral decomposition on its support,

$$J = \sum_{s=1}^{r} \lambda_s |e_s\rangle\langle e_s|, \qquad \lambda_s > 0.$$

Every v_k, w_j lies in this support: for $z \in \ker J$, $0 = \langle z|J|z\rangle = \sum_k |\langle z \mid v_k\rangle|^2$, and likewise for the w_j.

In the Kraus-label space $\mathbb{C}^N$, define

$$|a_s\rangle = \frac{1}{\sqrt{\lambda_s}} \sum_k \langle e_s \mid v_k\rangle |k\rangle, \qquad |b_s\rangle = \frac{1}{\sqrt{\lambda_s}} \sum_j \langle e_s \mid w_j\rangle |j\rangle.$$

Both lists are orthonormal, since

$$\langle a_s \mid a_t\rangle = \frac{\langle e_t|J|e_s\rangle}{\sqrt{\lambda_s \lambda_t}} = \delta_{st}, \qquad \langle b_s \mid b_t\rangle = \delta_{st}.$$

Complete each list to an orthonormal basis of $\mathbb{C}^N$ and let u be the unitary mapping the first basis to the second. In particular, $u|a_s\rangle = |b_s\rangle$ for $1 \le s \le r$. Writing $a_{ks} = \langle k \mid a_s\rangle$ and $b_{js} = \langle j \mid b_s\rangle$, we have $b_{js} = \sum_k u_{jk} a_{ks}$. Expanding in the support basis gives

$$|w_j\rangle = \sum_s \sqrt{\lambda_s}\, b_{js} |e_s\rangle = \sum_k u_{jk} \sum_s \sqrt{\lambda_s}\, a_{ks} |e_s\rangle = \sum_k u_{jk} |v_k\rangle.$$

Finally, the coefficients of $|i\rangle_R$ in these vectors are $L_j|i\rangle_A$ and $K_k|i\rangle_A$. Comparing them for every i proves $L_j = \sum_k u_{jk} K_k$. □

Example 4.22 (Discarding a subsystem is a channel)**.** For an orthonormal basis $|j\rangle$ of $\mathcal{H}_E$, let $K_j = I_A \otimes \langle j| \colon \mathcal{H}_A \otimes \mathcal{H}_E \to \mathcal{H}_A$. Then

$$\mathrm{tr}_E(W) = \sum_j K_j W K_j^\dagger, \qquad \sum_j K_j^\dagger K_j = I_A \otimes \sum_j |j\rangle\langle j| = I_{AE}.$$

Hence partial trace is CPTP. The full trace is the special case with one-dimensional output: its Kraus operators are the basis bras $\langle j|$. Although the output has no remaining quantum degrees of freedom, this is still a trace-preserving map to the scalars.

4.5 Measurements as Quantum Operations

The single-operator update $M_x \rho M_x^\dagger$ does not describe every possible measurement branch. An apparatus may record an outcome x while leaving additional information in an inaccessible environment. For a fixed recorded outcome, we must then sum over that information.

Definition 4.23 (Quantum instrument)**.** A *quantum instrument* with finite outcome set Ω is a family of quantum operations $\{\mathcal{I}_x : x \in \Omega\}$ with common input and output spaces such that $\sum_x \mathcal{I}_x$ is a channel. On input ρ, it gives outcome x with probability $p(x) = \mathrm{tr}[\mathcal{I}_x(\rho)]$ and conditional state $\mathcal{I}_x(\rho)/p(x)$ whenever $p(x) > 0$.

By Theorem 4.18, we can write

$$\mathcal{I}_x(\rho) = \sum_\alpha K_{x\alpha} \rho K_{x\alpha}^\dagger, \qquad \sum_{x,\alpha} K_{x\alpha}^\dagger K_{x\alpha} = I_A. \tag{4.12}$$

Its POVM is

$$\mu(x) = \sum_\alpha K_{x\alpha}^\dagger K_{x\alpha},$$

so $p(x) = \mathrm{tr}[\mu(x)\rho]$. The label α distinguishes unrecorded alternatives within outcome x. Discarding x gives the channel $\sum_x \mathcal{I}_x$. Keeping both the classical record and the quantum output gives the channel

$$\rho \longmapsto \sum_x |x\rangle\langle x|_C \otimes \mathcal{I}_x(\rho).$$

Its Kraus operators are $|x\rangle_C \otimes K_{x\alpha}$. The trace of each block is the probability of that recorded outcome.

Theorem 4.24 (Realizing an instrument)**.** *Every finite quantum instrument can be realized by adjoining a system in a fixed pure state, applying a unitary, measuring an auxiliary system projectively, and discarding auxiliary systems.*

Proof. Choose the Kraus operators in Eq. (4.12) and an environment basis $|x, \alpha\rangle$. Define

$$V|\psi\rangle = \sum_{x,\alpha} K_{x\alpha}|\psi\rangle \otimes |x, \alpha\rangle.$$

Completeness makes V an isometry. By Section 4.3, it can be implemented by a unitary with auxiliary systems in fixed pure states. Measure the environment with projections $P_x = \sum_\alpha |x, \alpha\rangle\langle x, \alpha|$. The unnormalized joint state for outcome x is

$$(I_B \otimes P_x) V \rho V^\dagger (I_B \otimes P_x).$$

Expanding V and tracing out the environment leaves $\sum_\alpha K_{x\alpha}\rho K_{x\alpha}^\dagger = \mathcal{I}_x(\rho)$. Its trace and normalization give exactly the required outcome probability and conditional state. □

For a single operator per outcome, the construction reads $V|\psi\rangle = \sum_x M_x|\psi\rangle \otimes |x\rangle$. Measuring the auxiliary register produces $M_x|\psi\rangle \otimes |x\rangle$ before normalization. Taking $M_x = \sqrt{\mu(x)}$ proves that every POVM can be realized this way.

Corollary 4.25 (Projective realization with a common embedding)**.** *For a finite family of POVMs $\{\mu^s(x)\}_x$ on $\mathcal{H}$, there is a larger space $\mathcal{K}$, one isometry $W\colon \mathcal{H} \to \mathcal{K}$, and a projective measurement $\{Q_x^s\}_x$ on $\mathcal{K}$ for each choice s, such that*

$$W^\dagger Q_x^s W = \mu^s(x).$$

In particular, the embedding of the input state is independent of s.

Proof. Pad the outcome sets with zero effects so that every POVM has n outcomes. Set $\mathcal{K} = \mathcal{H} \otimes \mathbb{C}^n$ and $W|\psi\rangle = |\psi\rangle \otimes |0\rangle$. For each s, the map

$$W|\psi\rangle \longmapsto \sum_x \sqrt{\mu^s(x)}|\psi\rangle \otimes |x\rangle$$

preserves inner products, and therefore extends to a unitary U_s on $\mathcal{K}$. Define

$$Q_x^s = U_s^\dagger (I \otimes |x\rangle\langle x|) U_s.$$

These are mutually orthogonal projections summing to $I_\mathcal{K}$. For any $|\psi\rangle, |\phi\rangle$,

$$\langle\phi|W^\dagger Q_x^s W|\psi\rangle = \langle\phi|\sqrt{\mu^s(x)}\sqrt{\mu^s(x)}|\psi\rangle = \langle\phi|\mu^s(x)|\psi\rangle.$$

This proves the compression identity with the same W for every s. □

This is the projective-measurement reduction used in Chapter 3. Applying W_A and W_B to a shared state preserves all joint probabilities, since

$$(W_A \otimes W_B)^\dagger (Q_a^x \otimes Q_b^y)(W_A \otimes W_B) = \mu_A^x(a) \otimes \mu_B^y(b).$$

No dependence of the embedded state on the questions is needed.

Proposition 4.26 (Local channels do not change the distant marginal)**.** *Let ρ_{AB} be a joint state and let $\mathcal{E}_A$ be a channel from A to A'. Then*

$$\operatorname{tr}_{A'}[(\mathcal{E}_A \otimes \mathrm{id}_B)(\rho_{AB})] = \rho_B.$$

Proof. Let $K_k\colon A \to A'$ be Kraus operators of $\mathcal{E}_A$. For every operator B on the second system, cyclicity of the full trace gives

$$\begin{aligned}&\operatorname{tr}\big[(I_{A'} \otimes B)(\mathcal{E}_A \otimes \mathrm{id}_B)(\rho_{AB})\big] \\ &\quad = \sum_k \operatorname{tr}\big[(K_k^\dagger K_k \otimes B)\rho_{AB}\big] = \operatorname{tr}\big[(I_A \otimes B)\rho_{AB}\big].\end{aligned}$$

The two reduced operators have the same trace against every B and therefore are equal. □

Conditioning on one outcome of a local instrument can change the conditional state assigned to the distant system. The proposition applies to the trace-preserving sum over all outcomes. This extends the measurement-statistics argument from Section 2.2 to arbitrary local channels.

4.6 Qubit Channels and Noise

Recall that every qubit state has the form

$$\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z) = \frac{1}{2}\begin{pmatrix} 1+r_z & r_x - ir_y \\ r_x + ir_y & 1-r_z \end{pmatrix}, \qquad \|r\| \le 1.$$

A qubit channel acts on its Bloch vector by an affine transformation

$$r \longmapsto Mr + c, \tag{4.13}$$

where M is a real 3×3 matrix and $c \in \mathbb{R}^3$. Indeed, put $\sigma_1 = X, \sigma_2 = Y, \sigma_3 = Z$. Since the output coordinates are $r'_i = \operatorname{tr}[\sigma_i \mathcal{E}(\rho)]$, linearity gives

$$M_{ij} = \tfrac{1}{2}\operatorname{tr}[\sigma_i \mathcal{E}(\sigma_j)], \qquad c_i = \tfrac{1}{2}\operatorname{tr}[\sigma_i \mathcal{E}(I)].$$

These numbers are real because the operators inside each trace are products of Hermitian operators: for Hermitian A, B, $\overline{\operatorname{tr}[AB]} = \operatorname{tr}[BA] = \operatorname{tr}[AB]$. These formulas determine M, c directly from the channel's action on I, X, Y, Z.

Why the image is an ellipsoid. The vector c is the image of the origin, hence the Bloch vector of $\mathcal{E}(I/2)$. To understand the effect of M, choose an orthonormal eigenbasis v_1, v_2, v_3 of the real symmetric positive matrix $M^T M$:

$$M^T M v_j = s_j^2 v_j, \qquad s_j \ge 0.$$

The numbers s_j are the *singular values* of M. For $s_j > 0$, put $u_j = Mv_j / s_j$. These vectors are orthonormal, since

$$u_i^T u_j = \frac{v_i^T M^T M v_j}{s_i s_j} = \frac{s_j^2}{s_i s_j}\, v_i^T v_j = \delta_{ij}.$$

If some s_j vanish, complete the already defined u_j to an orthonormal basis arbitrarily. In that case $Mv_j = 0$, because $\|Mv_j\|^2 = v_j^T M^T M v_j = 0$. Thus $Mv_j = s_j u_j$ for every j. Equivalently, if Q and R have columns u_j and v_j, respectively,

$$M = Q\,\operatorname{diag}(s_1, s_2, s_3) R^T, \qquad Q^T Q = R^T R = I.$$

This constructs the real singular value decomposition directly from the spectral theorem.

Write an input vector as $r = \sum_j x_j v_j$. Since the v_j are orthonormal, $\|r\|^2 = \sum_j x_j^2$, and

$$Mr + c = c + \sum_j s_j x_j u_j.$$

Therefore the output set is centered at c, has principal directions u_j, and has semiaxis lengths s_j. If every $s_j > 0$, it is the solid ellipsoid

$$\left\{ r' : \sum_{j=1}^{3} \frac{\left(u_j^T (r' - c)\right)^2}{s_j^2} \le 1 \right\}.$$

Indeed, setting $y_j = u_j^T (r' - c)$ gives $y_j = s_j x_j$, so the displayed inequality is exactly $\sum_j x_j^2 \le 1$. When a singular value vanishes, the corresponding output coordinate y_j must be zero; sum

only over the positive singular values. Ranks two, one, and zero therefore give a filled ellipse, a segment, and a point, respectively.

Geometrically, R^T changes to the input principal axes, the diagonal matrix scales their coordinates, Q places the result along the output principal axes, and c translates its center. Orthogonal changes of axes can include reflections. The individual factors need not themselves represent physical quantum channels.

Since the whole image lies inside the Bloch ball, each $s_j \le 1$: the two allowed input vectors $\pm v_j$ give the outputs $c \pm s_j u_j$, whose distance is $2s_j$ and cannot exceed the diameter 2 of the ball. This bound alone does not ensure that a translated ellipsoid fits in the ball. Moreover, even mapping the entire ball into itself guarantees only positivity on single-qubit states, not complete positivity. Transposition gives the reflection $(r_x, r_y, r_z) \mapsto (r_x, -r_y, r_z)$, which preserves the ball but fails the reference-system test in Example 4.16.

Exercise 4.27 (Unitary channels rotate the Bloch sphere)**.** Show that conjugation by a qubit unitary acts on Bloch vectors as a rotation about the origin.

Solution. For $\mathcal{E}(A) = UAU^\dagger$, we have $\mathcal{E}(I) = I$, so $c = 0$. The traceless Hermitian matrices $U\sigma_j U^\dagger$ expand as $\sum_i M_{ij}\sigma_i$. Using $\mathrm{tr}[\sigma_i\sigma_j] = 2\delta_{ij}$ and cyclicity of the trace,

$$2\delta_{jk} = \mathrm{tr}[(U\sigma_j U^\dagger)(U\sigma_k U^\dagger)] = 2\sum_i M_{ij}M_{ik}.$$

Thus $M^T M = I$. To show that the determinant is $+1$, write $U = e^{iK}$ as in Exercise 4.2. The path $U(t) = e^{itK}$, $0 \le t \le 1$, gives a continuous path of orthogonal matrices $M(t)$. Their determinants belong to $\{1, -1\}$ and cannot change continuously between these two values. Since $M(0) = I$, $\det M(1) = 1$. Therefore M is a rotation.

Conversely, a rotation through angle θ about a unit vector n is realized by

$$U = \exp(-i\theta\, n\cdot\sigma/2) = \cos(\theta/2)I - i\sin(\theta/2)n\cdot\sigma.$$

Using $(a\cdot\sigma)(b\cdot\sigma) = (a\cdot b)I + i(a\times b)\cdot\sigma$ gives

$$r' = \cos\theta\, r + (1-\cos\theta)(n\cdot r)n + \sin\theta(n\times r),$$

the usual rotation formula. An overall phase multiplying U has no effect on the channel. □

4.6.1 The Bit-Flip Channel

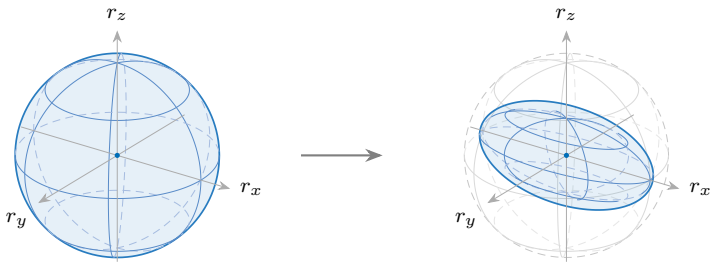
For the Pauli channels, we use p for the probability of applying the indicated Pauli operator. The *bit-flip channel* is

$$\mathcal{B}_p(\rho) = (1-p)\rho + pX\rho X, \qquad 0 \le p \le 1. \tag{4.14}$$

Its Kraus operators are $\sqrt{1-p}I$ and $\sqrt{p}X$. Since $XXX = X$, $XYX = -Y$, and $XZX = -Z$, its Bloch action is

$$(r_x, r_y, r_z) \longmapsto (r_x, (1-2p)r_y, (1-2p)r_z).$$

The x-axis is fixed and the two perpendicular coordinates contract in magnitude by $|1-2p|$. For $p > 1/2$ they also change sign. At $p = 1$ the channel is unitary, so there is no contraction. At $p = 1/2$ those coordinates vanish and the whole ball maps to the x-axis segment.



Input Bit flip ($p = 0.3$)

Figure 4.1: Bit flip with $p = 0.3$. The x-axis is fixed; both transverse semiaxes shrink to 0.4. The blue surface bounds the image of the entire Bloch ball. The grey wireframe marks the original sphere, and the blue dot marks the center. All channel figures use the same three-dimensional viewpoint.

Exercise 4.28 (An unread measurement in the X-basis)**.** Let $P_+ = |+\rangle\langle+|$ and $P_- = |-\rangle\langle-|$. Determine the Bloch-vector action of $\mathcal{E}(\rho) = P_+\rho P_+ + P_-\rho P_-$ and describe its image.

Solution. Since $P_\pm = (I \pm X)/2$,

$$P_+\rho P_+ + P_-\rho P_- = \tfrac{1}{4}((I+X)\rho(I+X) + (I-X)\rho(I-X)) = \tfrac{1}{2}(\rho + X\rho X).$$

This is the bit-flip channel with $p = 1/2$, so $(r_x, r_y, r_z) \mapsto (r_x, 0, 0)$. The Bloch ball is mapped onto its x-axis segment. Its endpoints, the states $|+\rangle$ and $|-\rangle$, are unchanged. □

4.6.2 The Phase-Flip Channel

The *phase-flip channel* applies Z with probability p:

$$\mathcal{Z}_p(\rho) = (1-p)\rho + pZ\rho Z, \qquad 0 \le p \le 1.$$

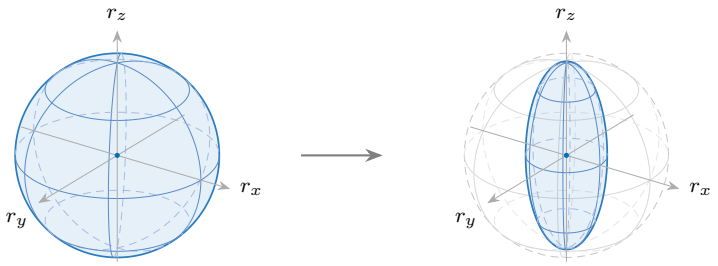
Its Kraus operators are $\sqrt{1-p}I$ and $\sqrt{p}Z$. Conjugation by Z changes the signs of X, Y and fixes Z, so

$$(r_x, r_y, r_z) \longmapsto ((1-2p)r_x, (1-2p)r_y, r_z).$$

Thus every state on the z-axis is fixed, while the transverse coordinates contract in magnitude by $|1-2p|$. For $p = 1/2$,

$$\tfrac{1}{2}(\rho + Z\rho Z) = P_0\rho P_0 + P_1\rho P_1,$$

which is complete dephasing in the computational basis. The same map has appeared as an unread measurement, a discarded environment, and a random unitary operation.



Input Phase flip ($p = 0.3$)

Figure 4.2: Phase flip with $p = 0.3$. The z-axis is fixed and the two transverse semiaxes shrink to 0.4. At $p = 1/2$, the output ellipsoid collapses to the z-axis segment.

4.6.3 The Bit–Phase-Flip Channel

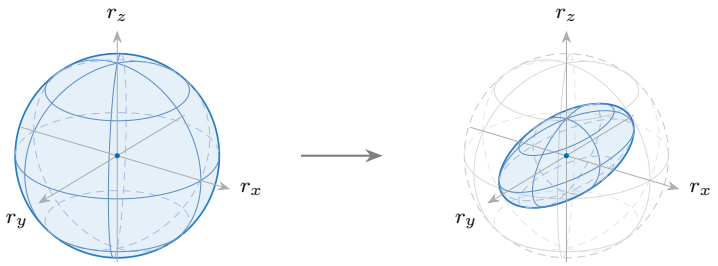
Since $Y = iXZ$, conjugation by Y combines a bit flip and a phase flip; the overall factor i has no effect on the density operator. The channel

$$\mathcal{Y}_p(\rho) = (1-p)\rho + pY\rho Y, \qquad 0 \le p \le 1,$$

has Kraus operators $\sqrt{1-p}I$ and $\sqrt{p}Y$. Its Bloch action is

$$(r_x, r_y, r_z) \longmapsto ((1-2p)r_x, r_y, (1-2p)r_z).$$

It fixes the y-axis and contracts the perpendicular coordinates. The three Pauli channels have the same geometric form, with different fixed axes.



Input Bit–phase flip ($p = 0.3$)

Figure 4.3: Bit–phase flip with $p = 0.3$. The y-axis is fixed, while the x- and z-semiaxes shrink to 0.4. The common viewpoint makes the different fixed axes of the three Pauli channels visible.

4.6.4 The Depolarizing Channel

The *depolarizing channel* replaces the qubit by $I/2$ with probability p and leaves it unchanged with probability $1-p$:

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I}{2}, \qquad 0 \le p \le 1. \tag{4.15}$$

Its linear extension to arbitrary operators is $\mathcal{D}_p(A) = (1-p)A + p\,\mathrm{tr}[A]I/2$. The trace factor is necessary when A is not a density operator. In Bloch coordinates,

$$r \longmapsto (1-p)r.$$

All directions contract equally. The output is a ball of radius $1-p$ centered at the origin; at $p=1$ it is the single point corresponding to $I/2$.

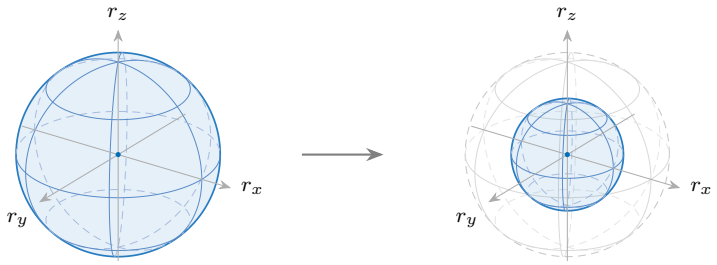


Figure 4.4: Depolarization with $p = 0.45$. The Bloch ball contracts uniformly by the factor $1-p = 0.55$; its center stays at the maximally mixed state.

The following calculation supplies a Kraus representation and explains how Pauli noise realizes this channel.

Exercise 4.29 (Averaging over Pauli operators)**.** Define

$$\mathcal{T}(A) = \tfrac{1}{4}(A + XAX + YAY + ZAZ).$$

Compute $\mathcal{T}(I), \mathcal{T}(X), \mathcal{T}(Y), \mathcal{T}(Z)$ and deduce that $\mathcal{T}(\rho) = I/2$ for every qubit density operator ρ.

Solution. All four conjugations fix I, so $\mathcal{T}(I) = I$. For X, conjugation by I and X gives X, whereas conjugation by Y and Z gives $-X$. Thus $\mathcal{T}(X) = 0$. The same argument gives $\mathcal{T}(Y) = \mathcal{T}(Z) = 0$. Expanding $\rho = (I + r_x X + r_y Y + r_z Z)/2$ proves the claim. More generally, $\mathcal{T}(A) = \mathrm{tr}[A]I/2$ for every operator A. □

Substituting the Pauli average into Eq. (4.15),

$$\mathcal{D}_p(\rho) = \left(1 - \frac{3p}{4}\right)\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z).$$

Its Kraus operators may therefore be chosen as

$$\sqrt{1-3p/4}\, I, \qquad \sqrt{p/4}\, X, \qquad \sqrt{p/4}\, Y, \qquad \sqrt{p/4}\, Z.$$

Here p is the replacement probability; the probability of a nonidentity Pauli in this representation is $3p/4$. These are different parameterizations of the same family, so their parameters should not be interchanged.

Exercise 4.30 (Depolarization in dimension d)**.** Find Kraus operators for the channel

$$\mathcal{D}_p(\rho) = (1-p)\rho + pI/d$$

on a d-dimensional system, where $0 \le p \le 1$.

Solution. Choose an orthonormal basis $|0\rangle, \dots, |d-1\rangle$ and set

$$K_* = \sqrt{1-p}\, I, \qquad K_{ij} = \sqrt{p/d}\, |i\rangle\langle j| \quad (0 \le i, j < d).$$

The second part of the operator sum is

$$\sum_{i,j} K_{ij} A K_{ij}^\dagger = \frac{p}{d} \sum_{i,j} \langle j|A|j\rangle |i\rangle\langle i| = p \operatorname{tr}[A] \frac{I}{d}.$$

Also,

$$K_*^\dagger K_* + \sum_{i,j} K_{ij}^\dagger K_{ij} = (1-p)I + \frac{p}{d} \sum_{i,j} |j\rangle\langle j| = I.$$

Thus these operators realize the desired channel. This representation has d^2+1 listed operators and need not be minimal; the rank bound in Corollary 4.19 guarantees that a list of at most d^2 nonzero operators is possible. □

4.6.5 Amplitude Damping

Random Pauli operations are useful models of noise, but they do not describe all channels. A system may preferentially lose energy to its environment, or lose coherence while its energy populations stay fixed. We now compare these two mechanisms.

Let $|0\rangle$ be the ground state and $|1\rangle$ an excited state. Suppose an excitation is transferred to an initially unexcited environment with probability γ. A model isometry, with system first and environment second, is

$$V|0\rangle = |00\rangle, \qquad V|1\rangle = \sqrt{1-\gamma}|10\rangle + \sqrt{\gamma}|01\rangle, \qquad 0 \le \gamma \le 1. \tag{4.16}$$

The two displayed vectors are orthonormal, so this map has a unitary realization. Taking its environment matrix elements gives

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}. \tag{4.17}$$

The *amplitude-damping channel* is $\mathcal{A}_\gamma(\rho) = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger$. The K_1 branch transfers an excitation to the environment. The K_0 branch is not generally the identity: conditional on seeing no excitation in the environment, the relative weight of $|1\rangle$ is reduced. The probability of the transfer branch is $\gamma\langle 1|\rho|1\rangle$, which depends on the input.

Exercise 4.31 (Amplitude damping of a density matrix)**.** For $\rho = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$ with $a + c = 1$, calculate $\mathcal{A}_\gamma(\rho)$ using Eq. (4.17).

Solution. The two terms are

$$K_0 \rho K_0^\dagger = \begin{pmatrix} a & \sqrt{1-\gamma}\, b \\ \sqrt{1-\gamma}\, \bar{b} & (1-\gamma)c \end{pmatrix},$$

$$K_1 \rho K_1^\dagger = \begin{pmatrix} \gamma c & 0 \\ 0 & 0 \end{pmatrix}.$$

Consequently,

$$\mathcal{A}_\gamma(\rho) = \begin{pmatrix} a + \gamma c & \sqrt{1-\gamma}\, b \\ \sqrt{1-\gamma}\, \overline{b} & (1-\gamma)c \end{pmatrix}. \tag{4.18}$$

The excited population decreases from c to $(1-\gamma)c$ and the ground-state population gains exactly the amount γc. The off-diagonal entries are multiplied by $\sqrt{1-\gamma}$. □

In Bloch coordinates the same calculation gives

$$(r_x, r_y, r_z) \longmapsto (\sqrt{1-\gamma}\, r_x, \sqrt{1-\gamma}\, r_y, \gamma + (1-\gamma) r_z). \tag{4.19}$$

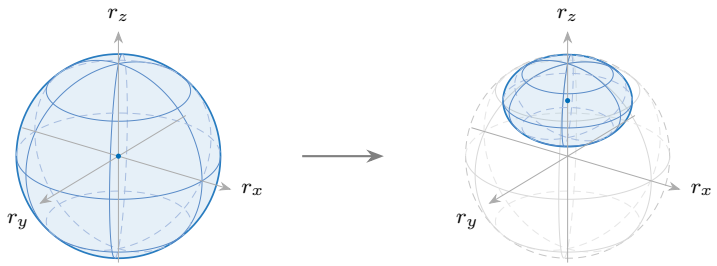
The affine parameters are

$$M = \operatorname{diag}(\sqrt{1-\gamma}, \sqrt{1-\gamma}, 1-\gamma), \qquad c = \begin{pmatrix} 0 \\ 0 \\ \gamma \end{pmatrix}.$$

Thus the center of the output ellipsoid is $(0, 0, \gamma)$, its two transverse semiaxes have length $\sqrt{1-\gamma}$, and its vertical semiaxis has length $1-\gamma$. For $0 \le \gamma < 1$, its equation is

$$\frac{(r'_x)^2 + (r'_y)^2}{1-\gamma} + \frac{(r'_z - \gamma)^2}{(1-\gamma)^2} \le 1.$$

Its upper endpoint is always the north pole $(0, 0, 1)$, while its lower endpoint is $(0, 0, 2\gamma - 1)$. At $\gamma = 0$ the image is the whole ball; at $\gamma = 1$ all semiaxes vanish and the image is the single point $(0, 0, 1)$. This is the displacement and contraction shown in Fig. 4.5.



Input Amplitude damping ($\gamma = 0.6$)

Figure 4.5: Amplitude damping with $\gamma = 0.6$. The image has center $(0, 0, 0.6)$, transverse semiaxes $\sqrt{0.4}$, and vertical semiaxis 0.4. It touches the original sphere at the north pole $|0\rangle$. The center shifts upward as the semiaxes shrink.

For $\gamma > 0$, the only fixed density operator is $|0\rangle\langle 0|$: Eq. (4.18) forces $c = 0$ and $b = 0$. At $\gamma = 1$, every state is reset to $|0\rangle\langle 0|$. If the energy of $|0\rangle$ is set to zero and that of $|1\rangle$ to $\Delta > 0$, the expected energy decreases from Δc to $\Delta(1-\gamma)c$.

Example 4.32 (The Choi operator of amplitude damping)**.** The two Kraus operators correspond to the vectors

$$|v_0\rangle = |00\rangle + \sqrt{1-\gamma}|11\rangle, \qquad |v_1\rangle = \sqrt{\gamma}|10\rangle.$$

With the reference factor first and the standard product-basis order,

$$J_{\mathcal{A}_\gamma} = \begin{pmatrix} 1 & 0 & 0 & \sqrt{1-\gamma} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma & 0 \\ \sqrt{1-\gamma} & 0 & 0 & 1-\gamma \end{pmatrix}.$$

It is positive as a sum of two positive rank-one operators, and its output partial trace is I. For $\gamma > 0$ its rank is two, so the two Kraus operators are minimal. At $\gamma = 0$ it is the Choi operator of the identity channel and has rank one.

4.6.6 Phase Damping

Coherence can disappear without any change in the two energy populations. Consider the isometry

$$V|0\rangle = |0\rangle|0\rangle, \qquad V|1\rangle = |1\rangle\big(\sqrt{1-\lambda}|0\rangle + \sqrt{\lambda}|1\rangle\big), \qquad 0 \le \lambda \le 1.$$

The system stays in its original basis state, while the environment acquires information about that state.

Discarding the environment produces the *phase-damping channel* $\mathcal{P}_\lambda$ with Kraus operators

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \qquad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}. \tag{4.20}$$

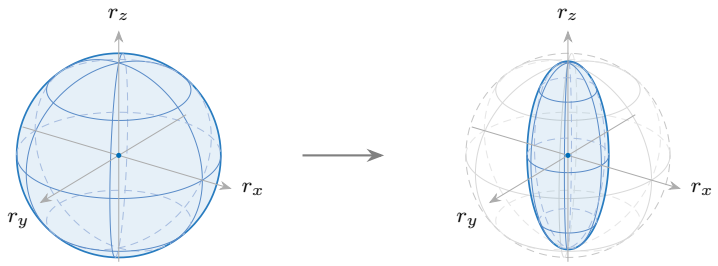
For a general density matrix,

$$\mathcal{P}_\lambda \begin{pmatrix} a & b \\ \overline{b} & 1-a \end{pmatrix} = \begin{pmatrix} a & \sqrt{1-\lambda}\, b \\ \sqrt{1-\lambda}\, \overline{b} & 1-a \end{pmatrix}. \tag{4.21}$$

The diagonal entries remain fixed, so the expectation of any Hamiltonian diagonal in this basis remains fixed. The off-diagonal entries are reduced by the overlap $\sqrt{1-\lambda}$ of the two environment states. At $\lambda = 1$ they are orthogonal and all coherence in this basis disappears.

In Bloch coordinates,

$$(r_x, r_y, r_z) \longmapsto (\sqrt{1-\lambda}\, r_x, \sqrt{1-\lambda}\, r_y, r_z).$$



Input Phase damping ($\lambda = 0.84$)

Figure 4.6: Phase damping with $\lambda = 0.84$. The transverse semiaxes shrink to $\sqrt{1-\lambda} = 0.4$, while the z-axis stays fixed. The center and the populations in the computational basis are unchanged.

Comparing with the phase-flip channel shows that

$$\mathcal{P}_\lambda(\rho) = (1-p)\rho + pZ\rho Z, \qquad p = \frac{1-\sqrt{1-\lambda}}{2}. \tag{4.22}$$

Here $0 \le p \le 1/2$. This is an equality of the reduced dynamics, even though the two constructions describe the environment differently. The remainder of the phase-flip family, with $p > 1/2$, additionally reverses the signs of both transverse Bloch coordinates.

Exercise 4.33 (Two Kraus descriptions of phase damping)**.** Find a 2×2 unitary matrix u which transforms the Kraus operators E_0, E_1 in Eq. (4.20) into $F_0 = \sqrt{1-p}I$ and $F_1 = \sqrt{p}Z$, with p as in Eq. (4.22): $F_j = \sum_k u_{jk}E_k$.

Solution. Write $a = \sqrt{1-p}$, $b = \sqrt{p}$, and $s = \sqrt{1-\lambda} = 1 - 2p = a^2 - b^2$. Then $\sqrt{\lambda} = 2ab$. Choose

$$u = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

Its rows are orthonormal since $a^2 + b^2 = 1$, so it is unitary. Direct calculation gives

$$\begin{aligned} aE_0 + bE_1 &= \operatorname{diag}(a, as + 2ab^2) = aI, \\ bE_0 - aE_1 &= \operatorname{diag}(b, bs - 2a^2 b) = bZ. \end{aligned}$$

The calculation also covers $\lambda = 0$ and $\lambda = 1$. Thus the two representations are related exactly as in Theorem 4.21. □

4.6.7 Unitality and the Scope of the Model

A channel from a system to itself is *unital* if $\mathcal{E}(I) = I$. Equivalently, it preserves the maximally mixed state. In the qubit Bloch representation, this means $c = 0$ in Eq. (4.13). Every random unitary channel is unital, since $\sum_k p_k U_k I U_k^\dagger = I$.

Exercise 4.34 (Which noise channels are unital?)**.** Show that depolarization and phase damping are unital. Show that amplitude damping is not unital for $\gamma > 0$.

Solution. The linear extension of depolarization satisfies

$$\mathcal{D}_p(I) = (1-p)I + p\,\mathrm{tr}[I]I/2 = I.$$

For phase damping, its diagonal Kraus operators give

$$E_0E_0^\dagger + E_1E_1^\dagger = \mathrm{diag}(1, 1-\lambda) + \mathrm{diag}(0, \lambda) = I.$$

For amplitude damping,

$$\mathcal{A}_\gamma(I) = K_0K_0^\dagger + K_1K_1^\dagger = \begin{pmatrix} 1+\gamma & 0 \\ 0 & 1-\gamma \end{pmatrix} = I + \gamma Z.$$

This differs from I whenever $\gamma > 0$. At $\gamma = 0$, amplitude damping is the identity channel and is unital. □

Consequently, nontrivial amplitude damping cannot be expressed as a classical random choice of unitary operations. It can even increase purity: at $\gamma = 1$, it sends $I/2$ to the pure state $|0\rangle\langle 0|$. Noise therefore need not mean movement towards the maximally mixed state. In this example the environment removes energy and carries away information about the input.

Scope of the description. The environment construction assumes a fixed initial environment state, independent of the input, and no initial system–environment correlations. If the preparation leaves information about the input in the environment and that environment subsequently interacts with the system, the reduced input state alone may not determine the output. There need not then be a single channel on all input states describing the entire preparation-dependent procedure. This does not contradict unitary dynamics of the combined system. It identifies which information must be included in its initial description.

A channel for one time interval does not, by itself, assume memoryless dynamics across several intervals. Composition represents a sequence of channels when each step has the specified action independently of unrecorded correlations left by earlier steps.

Summary of Chapter 4: Quantum Operations

- **Closed-system evolution.** A unitary acts as $\rho \mapsto U\rho U^\dagger$, preserving the eigenvalues of ρ. For a fixed Hamiltonian,
$$U(t) = e^{-itH/\hbar}, \qquad \dot{\rho} = -\frac{i}{\hbar}[H, \rho].$$
- **Quantum operations and channels.** An operation is a complex-linear, completely positive map that does not increase the trace of positive operators. Complete positivity means that $\mathrm{id}_R \otimes \mathcal{E}$ is positive for every reference system R. A channel is a trace-preserving operation. Positivity on the input system alone is insufficient, as transposition shows.
- **Operator-sum representation.** Complete positivity is equivalent to the existence of Kraus operators:
$$\mathcal{E}(A) = \sum_k K_k A K_k^\dagger.$$
The trace condition is $\sum_k K_k^\dagger K_k \leq I$ for an operation, with equality for a channel. Composing maps with Kraus lists (K_k) and (L_l) gives the list $(L_l K_k)$.
- **Environment and measurement models.** Every channel has an isometry $V|\psi\rangle = \sum_k K_k|\psi\rangle \otimes |k\rangle$ satisfying $\mathcal{E}(\rho) = \mathrm{tr}_E[V\rho V^\dagger]$. It can be implemented by a unitary with auxiliary systems initially in fixed pure states. Individual outcomes are obtained by also measuring an auxiliary system. Every finite instrument, and hence every POVM, has such a realization.
- **The Choi test.** With input reference first and $|\Omega\rangle = \sum_i |i\rangle_R |i\rangle_A$,
$$J_\mathcal{E} = (\mathrm{id}_R \otimes \mathcal{E})(|\Omega\rangle\langle\Omega|), \qquad \mathcal{E}(A) = \mathrm{tr}_R[(A^T \otimes I_B) J_\mathcal{E}].$$
The map is completely positive exactly when $J_\mathcal{E} \geq 0$. For a channel, $\mathrm{tr}_B J_\mathcal{E} = I_R$; for an operation, $\mathrm{tr}_B J_\mathcal{E} \leq I_R$. Reshaping eigenvectors multiplied by the square roots of their eigenvalues gives Kraus operators. The minimum number of nonzero Kraus operators is $\mathrm{rank}\, J_\mathcal{E} \leq d_A d_B$.
- **Freedom of representation.** Two Kraus lists give the same map exactly when, after padding with zeros, $L_j = \sum_k u_{jk} K_k$ for a unitary matrix u. Different environment descriptions can therefore produce the same reduced dynamics.
- **Measurements, probabilities, and residual states.** An instrument is a finite family of operations $(\mathcal{I}_x)$ whose sum is a channel. It gives
$$p(x) = \mathrm{tr}[\mathcal{I}_x(\rho)], \qquad \rho_x = \frac{\mathcal{I}_x(\rho)}{p(x)} \quad \text{when } p(x) > 0.$$
If $\mathcal{I}_x(\rho) = \sum_\alpha K_{x\alpha} \rho K_{x\alpha}^\dagger$, its POVM is $\mu(x) = \sum_\alpha K_{x\alpha}^\dagger K_{x\alpha}$. The POVM specifies probabilities but not residual states. The unnormalized maps respect mixtures; their normalized conditional updates generally do not. Discarding the outcome gives $\sum_x \mathcal{I}_x$.
- **Local channels and distant systems.** A local channel leaves the other system's marginal unchanged: $\mathrm{tr}_{A'}[(\mathcal{E}_A \otimes \mathrm{id}_B)(\rho_{AB})] = \rho_B$. Conditioning on a local outcome can change the conditional distant state.

Summary of Chapter 4: Qubit Channels

Geometry. For $\rho = (I + r \cdot \sigma)/2$, a qubit channel acts as

$$r \longmapsto Mr + c, \qquad M_{ij} = \tfrac{1}{2}\operatorname{tr}[\sigma_i \mathcal{E}(\sigma_j)], \qquad c_i = \tfrac{1}{2}\operatorname{tr}[\sigma_i \mathcal{E}(I)].$$

The image of the Bloch ball is an ellipsoid centered at c. Its semiaxes are the singular values of M, and their directions are the output principal axes. Zero singular values give lower-dimensional images. Mapping the ball into itself ensures positivity, but complete positivity still requires the Choi test.

The channels in this chapter. All parameters lie in $[0, 1]$. For the Pauli channels, p is the probability of applying the indicated Pauli operator; for depolarization, p is the probability of replacement by $I/2$. To shorten the table, set

$$a = \sqrt{1-\gamma}, \qquad b = \sqrt{1-\lambda}.$$

Channel	Kraus operators	Output Bloch vector
Bit flip	$\sqrt{1-p}\,I,\ \sqrt{p}\,X$	$\big(r_x, (1-2p)r_y, (1-2p)r_z\big)$
Phase flip	$\sqrt{1-p}\,I,\ \sqrt{p}\,Z$	$\big((1-2p)r_x, (1-2p)r_y, r_z\big)$
Bit–phase flip	$\sqrt{1-p}\,I,\ \sqrt{p}\,Y$	$\big((1-2p)r_x, r_y, (1-2p)r_z\big)$
Depolarization	$\sqrt{1-3p/4}\,I,\ \sqrt{p/4}\,X,$ $\sqrt{p/4}\,Y,\ \sqrt{p/4}\,Z$	$(1-p)(r_x, r_y, r_z)$
Amplitude damping	$\operatorname{diag}(1, a),\ \sqrt{\gamma}\lvert 0\rangle\langle 1\rvert$	$\big(ar_x, ar_y, \gamma + (1-\gamma)r_z\big)$
Phase damping	$\operatorname{diag}(1, b),\ \sqrt{\lambda}\lvert 1\rangle\langle 1\rvert$	(br_x, br_y, r_z)

- **Fixed axes and displacement.** Each Pauli channel fixes its Pauli axis and contracts the transverse coordinates in magnitude. At $p = 1/2$, its image is the fixed-axis segment. Depolarization contracts all directions equally. Amplitude damping shifts the center to $(0, 0, \gamma)$ and, for $\gamma > 0$, has the unique fixed state $|0\rangle\langle 0|$.
- **Two descriptions of dephasing.** Phase damping equals phase flip with $p = (1 - \sqrt{1-\lambda})/2$. Both preserve the computational-basis populations. At $\lambda = 1$, they give an unread computational-basis measurement. Their different Kraus lists describe the same channel.
- **Unitality.** A channel is unital when $\mathcal{E}(I) = I$; for qubits, this is $c = 0$. The Pauli, depolarizing, and phase-damping channels are unital. Amplitude damping is nonunital for $\gamma > 0$ and can increase purity: at $\gamma = 1$, every input is reset to $|0\rangle\langle 0|$.
- **Scope of the model.** The environment is initially fixed independently of the input and uncorrelated with it. Preparation-dependent environment information can prevent the reduced input state from determining a single channel. A channel for one time interval does not by itself imply memoryless dynamics over successive intervals.